

INTERNATIONAL STANDARD

**Connectors for electrical and electronic equipment – Product requirements –
Part 2-111: Circular connectors – Detail specification for power connectors with
M12 screw-locking**



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**Connectors for electrical and electronic equipment – Product requirements –
Part 2-111: Circular connectors – Detail specification for power connectors with
M12 screw-locking**

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INTERNATIONAL ELECTROTECHNICAL COMMISSION

**CONNECTORS FOR ELECTRICAL AND ELECTRONIC EQUIPMENT –
PRODUCT REQUIREMENTS –****Part 2-111: Circular connectors –
Detail specification for power connectors with M12 screw-locking**

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International Standard IEC 61076-2-111 has been prepared by subcommittee 48B: Electrical connectors, of IEC technical committee 48: Electrical connectors and mechanical structures for electrical and electronic equipment.

The text of this International Standard is based on the following documents:

FDIS	Report on voting
48B/2601/FDIS	48B/2616/RVD

Full information on the voting for the approval of this International Standard can be found in the report on voting indicated in the above table.

This document has been drafted in accordance with the ISO/IEC Directives, Part 2.

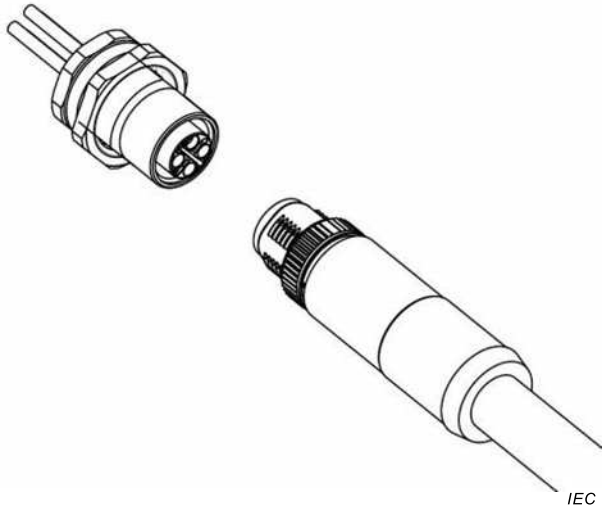
A list of all parts in the IEC 61076 series, published under the general title *Connectors for electrical and electronic equipment – Product requirements*, can be found on the IEC website.

Future standards in this series will carry the new general title as cited above. Titles of existing standards in this series will be updated at the time of the next edition.

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- reconfirmed,
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- amended.

A bilingual version of this publication may be issued at a later date.

IEC SC 48B – Electrical connectors Specification available from: IEC General secretariat Or from the addresses shown on the inside cover.	IEC 61076-2-111 Ed. 1
DETAIL SPECIFICATION in accordance with IEC 61076-1	
	Circular connectors Power connectors with M12 screw-locking Male and female connectors Male and female contacts Rewireable – Non-rewireable
	Free cable connectors Straight and right angle connectors Fixed connectors Flange mounting Single hole mounting

CONNECTORS FOR ELECTRICAL AND ELECTRONIC EQUIPMENT – PRODUCT REQUIREMENTS –

Part 2-111: Circular connectors – Detail specification for power connectors with M12 screw-locking

1 Scope

This part of IEC 61076-2 specifies 4 to 6-way circular connectors with M12 screw-locking with current ratings up to 16 A and voltage ratings of 63 V or 630 V, that are typically used for power supply and power applications in industrial premises. These connectors consist of both fixed and free connectors either rewirable or non-rewirable, with M12 screw-locking. Male connectors have round contacts Ø1,0 mm and Ø1,5 mm.

The different codings provided by this document prevent the mating of accordingly coded male or female connectors to any other similarly sized interfaces, covered by other standards and the cross-mating between the different codings provided by this document.

NOTE M12 is the dimension of the thread of the screw-locking mechanism of these circular connectors.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60050-581:2008, *International Electrotechnical Vocabulary – Part 581: Electromechanical components for electronic equipment*

IEC 60068-1, *Environmental testing – Part 1: General and guidance*

IEC 60068-2-60, *Environmental testing – Part 2-60: Tests – Test Ke: Flowing mixed gas corrosion test*

IEC 60352-2, *Solderless connections – Part 2: Crimped connections – General requirements, test methods and practical guidance*

IEC 60352-3, *Solderless connections – Part 3: Solderless accessible insulation displacement connections – General requirements, test methods and practical guidance*

IEC 60352-4, *Solderless connections – Part 4: Solderless non-accessible insulation displacement connections – General requirements, test methods and practical guidance*

IEC 60352-5, *Solderless connections – Part 5: Press-in connections – General requirements, test methods and practical guidance*

IEC 60352-6, *Solderless connections – Part 6: Insulation piercing connections – General requirements, test methods and practical guidance*

IEC 60352-7, *Solderless connections – Part 7: Spring clamp connections – General requirements, test methods and practical guidance*

IEC 60512-1-1, *Connectors for electronic equipment – Tests and measurements – Part 1-1: General examination – Test 1a: Visual examination*

IEC 60512-1-2, *Connectors for electronic equipment – Tests and measurements – Part 1-2: General examination – Test 1b: Examination of dimension and mass*

IEC 60512-2-1, *Connectors for electronic equipment – Tests and measurements – Part 2-1: Electrical continuity and contact resistance tests – Test 2a: Contact resistance – Millivolt level method*

IEC 60512-3-1, *Connectors for electronic equipment – Tests and measurements – Part 3-1: Insulation tests – Test 3a: Insulation resistance*

IEC 60512-4-1, *Connectors for electronic equipment – Tests and measurements – Part 4-1: Voltage stress tests – Test 4a: Voltage proof*

IEC 60512-5-1, *Connectors for electronic equipment – Tests and measurements – Part 5-1: Current-carrying capacity tests – Test 5a: Temperature rise*

IEC 60512-6-3, *Connectors for electronic equipment – Tests and measurements – Part 6-3: Dynamic stress tests – Test 6c: Shock*

IEC 60512-6-4, *Connectors for electronic equipment – Tests and measurements – Part 6-4: Dynamic stress tests – Test 6d: Vibration (sinusoidal)*

IEC 60512-9-1, *Connectors for electronic equipment – Tests and measurements – Part 9-1: Endurance tests – Test 9a: Mechanical operation*

IEC 60512-9-2, *Connectors for electronic equipment – Tests and measurements – Part 9-2: Endurance tests – Test 9b: Electrical load and temperature*

IEC 60512-11-1, *Electromechanical components for electronic equipment – Basic testing procedures and measuring methods – Part 11: Climatic tests – Section 1: Test 11a – Climatic sequence*

IEC 60512-11-4, *Connectors for electronic equipment – Tests and measurements – Part 11-4: Climatic tests – Test 11d: Rapid change of temperature*

IEC 60512-11-7, *Connectors for electronic equipment – Tests and measurements – Part 11-7: Climatic tests – Test 11g: Flowing mixed gas corrosion test*

IEC 60512-11-9, *Connectors for electronic equipment – Tests and measurements – Part 11-9: Climatic tests – Test 11i: Dry heat*

IEC 60512-11-10, *Connectors for electronic equipment – Tests and measurements – Part 11-10: Climatic tests – Test 11j: Cold*

IEC 60512-11-12, *Connectors for electronic equipment – Tests and measurements – Part 11-12: Climatic tests – Test 11m: Damp heat, cyclic*

IEC 60512-13-2, *Connectors for electronic equipment – Tests and measurements – Part 13-2: Mechanical operation tests – Test 13b: Insertion and withdrawal forces*

IEC 60512-13-5, *Connectors for electronic equipment – Tests and measurements – Part 13-5: Mechanical operation tests – Test 13e: Polarizing and keying method*

IEC 60512-16-5, *Connectors for electronic equipment – Tests and measurements – Part 16-5: Mechanical tests on contacts and terminations – Test 16e: Gauge retention force (resilient contacts)*

IEC 60512-19-3, *Electromechanical components for electronic equipment – Basic testing procedures and measuring methods – Part 19: Chemical resistance tests – Section 3: Test 19c – Fluid resistance*

IEC 60529:1989, *Degrees of protection provided by enclosures (IP Code)*

IEC 60529:1989/AMD1:1999

IEC 60529:1989/AMD2:2013

IEC 60664-1, *Insulation coordination for equipment within low-voltage systems – Part 1: Principles, requirements and tests*

IEC 60999-1, *Connecting devices – Electrical copper conductors – Safety requirements for screw-type and screwless-type clamping units – Part 1: General requirements and particular requirements for clamping units for conductors from 0,2 mm² up to 35 mm² (included)*

IEC 61076-1:2006, *Connectors for electronic equipment – Product requirements – Part 1: Generic specification*

IEC 61984, *Connectors – Safety requirements and tests*

IEC 62197-1, *Connectors for electronic equipment – Quality assessment requirements – Part 1: Generic specification*

ISO 1302, *Geometrical Product Specifications (GPS) – Indication of surface texture in technical product documentation*

ISO 11469, *Plastics – Generic identification and marking of plastics products*

3 Terms and definitions

For the purposes of this document, the terms and definitions given in IEC 61076-1, IEC 60050-581 as well as the following apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- IEC Electropedia: available at <http://www.electropedia.org/>
- ISO Online browsing platform: available at <http://www.iso.org/obp>

3.1

protective conductor

PE

conductor provided for purposes of safety, for example protection against electric shock

Note 1 to entry: In an electrical installation, the conductor identified PE is normally also considered as protective earthing conductor.

[SOURCE: IEC 60050-581:2008, 581-27-26]

3.2

functional earth conductor

FE

functional grounding conductor in US

earthing conductor provided for functional earthing

Note 1 to entry: Functional earthing a point or points in a system or in an installation or in equipment, for purposes other than electrical safety. [IEC 60050-195, Amendment 1: 2001, 195-01-13]

[SOURCE: IEC 60050-195:1998, 195-02-15]

3.3

mounting orientation

circular mounting position of the connector in relation to the polarization of the mating interface

Note 1 to entry: Where the free connector has an angled cable entry (as opposed to a straight cable entry), the angle between the cable entry direction and the polarization keyway should be specified.

4 Technical information

4.1 Systems of levels

4.1.1 Performance levels

Performance levels for these connectors are specified in Table 33.

4.1.2 Compatibility levels, according to IEC 61076-1

Connectors according to this document are intermateable according to IEC 61076-1.

4.2 Classification into climatic categories

Classification into climatic categories is specified in 6.3.

4.3 Creepage and clearance distances

Shall be as per 6.4.1 of this document.

4.4 Current-carrying capacity

Current carrying capacity as specified in Table 1 and 6.4.4.

Table 1 – Ratings of connectors

Coding	Style	No. of contacts	Function	pin Ø mm	Rated voltage V	Rated current A
E	5 way	3	2 + FE	1,5	63	16
		4	3 + FE			
		5	4 + FE			
F	4 way	2	2	1,5	63	16
		3	3			
		4	4			
K	5 way (4 + PE)	3	2 + PE	1,5	630	12
		4	3 + PE			
		5	4 + PE			
L	5 way	3	2 + FE	1,5	63	16
		4	3 + FE			
		5	4 + FE			
	4 way	2	2			
		3	3			
		4	4			
M	6 way (5 + PE)	3	2 + PE	1,0	630	8
		4	3 + PE			
		5	4 + PE			
		6	5 + PE			
S	4 way (3 + PE)	3	2 + PE	1,5	630	12
		4	3 + PE			
T	4 way	2	2	1,5	63	12
		3	3			
		4	4			

4.5 Marking

The marking of the connector and the package shall be in accordance with 2.7 of IEC 61076-1:2006.

4.6 Safety aspects

Some of the connector types covered by this document are only applicable for extra-low voltage (ELV) applications. For styles without a PE contact, IEC 61984 dealing with safety aspects of connectors is applicable as a guidance.

For K-coding, M-coding, and S-coding connectors IEC 61984 applies.

5 Dimensional information

5.1 General

Throughout this document, dimensions are in mm. Drawings are shown in the first angle projection. The shape of the connectors may deviate from those given in the following drawings as long as the specified dimensions are not influenced.

Missing dimensions shall be chosen according to common characteristics and intended use.

5.2 Survey of styles and variants

5.2.1 General

For all connector styles with cables, the length L of the cable shall be agreed between manufacturer and user. For interface dimensions see 5.3.

The interface dimensions of the female styles shall be chosen according to the common characteristics of the male styles.

For reliable intermateability, the dimensions of the female connector body as detailed in Annex A have to be met.

5.2.2 Contact terminations

The contact terminations shall be of the following types: screw, crimp, spring type, piercing, insulation displacement, press-in or solder.

5.2.3 Number of contacts or contact cavities

Table 2 shows the number of contacts or contact cavities.

Table 2 – Connector coding and number of contacts

Coding	No. of contact cavities	No. of contacts	Function
E	5	3	2 + FE
		4	3 + FE
		5	4 + FE
F	4	2	2
		3	3
		4	4
K	5	3	2 + PE
		4	3 + PE
		5	4 + PE
L	5	3	2 + FE
		4	3 + FE
		5	4 + FE
	4	2	2
		3	3
		4	4
M	6	3	2 + PE
		4	3 + PE
		5	4 + PE
		6	5 + PE
S	4	2	2 + PE
		3	3 + PE
T	4	2	2
		3	3
		4	4

5.2.4 Fixed connectors

5.2.4.1 General

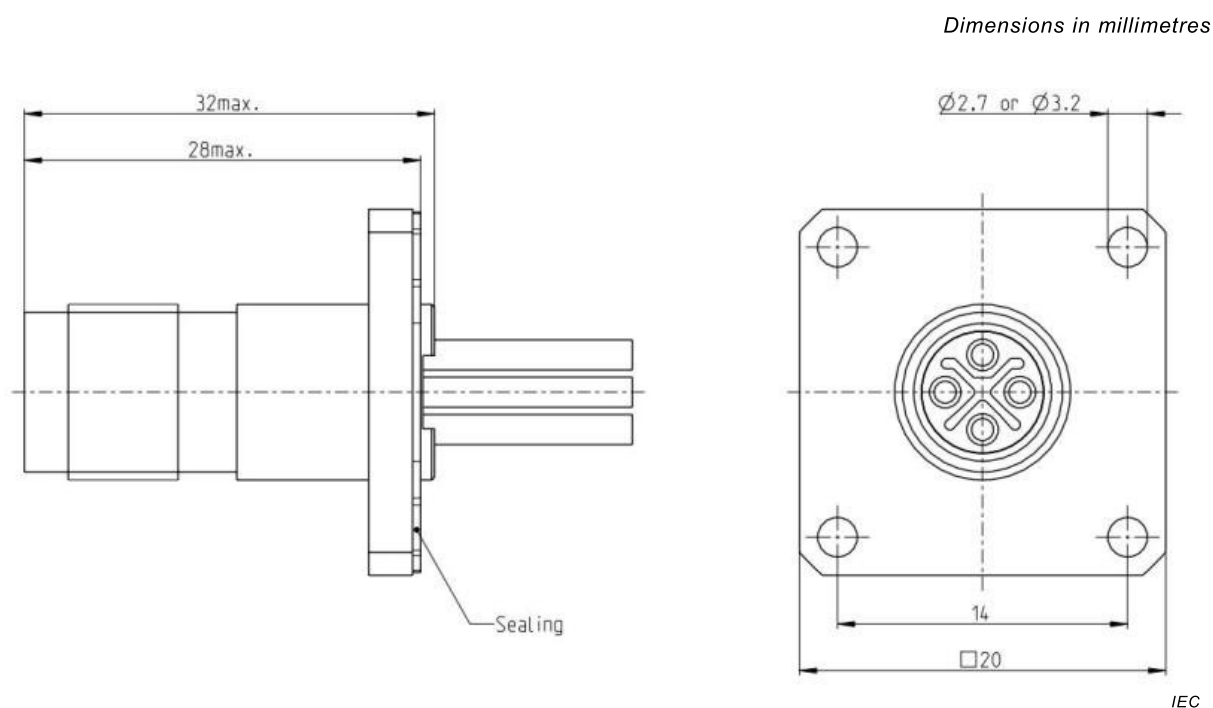
Table 3 shows styles of fixed connectors.

Table 3 – Styles of fixed connectors

Style	Description
DM	Fixed connector, male contacts, mounting with thread M12 × 1, square flange front mounting
EM	Fixed connector, male contacts, mounting with thread M12 × 1, with wire ends, single hole mounting M16 × 1,5
FM	Fixed connector, male contacts, mounting with thread M12 × 1, with wire ends, single hole mounting M20 × 1,5
GM	Fixed connector, male contacts, mounting with thread M12 × 1, with wire ends, single hole mounting M16 × 1,5, mounting orientation
HM	Fixed connector, male contacts, mounting with thread M12 × 1, with wire ends, single hole mounting M20 × 1,5, mounting orientation
IM	Fixed connector, male contacts, mounting with thread M12 × 1, with wire ends, single hole mounting M16 × 1,5, rear mounting
EF	Fixed connector, female contacts, with wire ends, single hole mounting M16 × 1,5
FF	Fixed connector, female contacts, with wire ends, single hole mounting M20 × 1,5
GF	Fixed connector, female contacts, with wire ends, single hole mounting M16 × 1,5, mounting orientation
HF	Fixed connector, female contacts, with wire ends, single hole mounting M20 × 1,5, mounting orientation
IF	Fixed connector, female contacts, with wire ends, single hole mounting M16 × 1,5, rear mounting

5.2.4.2 Style DM

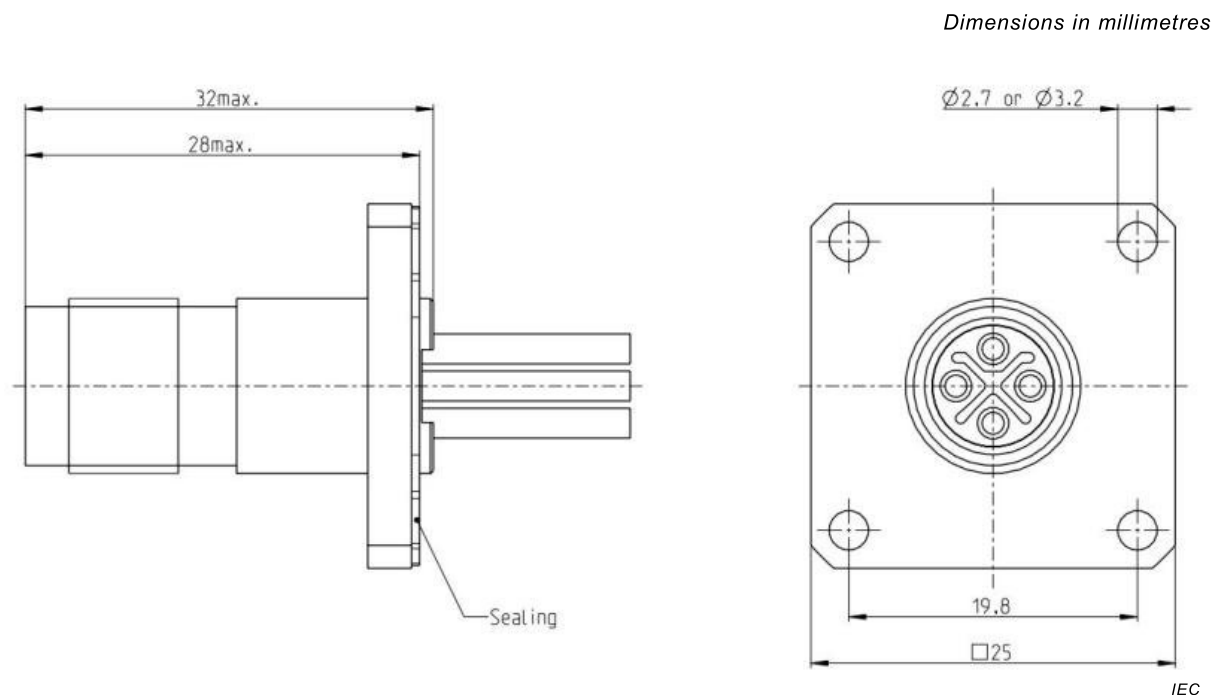
Figure 1 shows a fixed connector, with male contacts, mounting with thread M12 × 1 and a square flange 25 mm and front mounting.



NOTE Orientation of coding, see Annex A.

Figure 1 – Fixed connector, male contacts, mounting with thread M12 x 1, square flange 20 mm, front mounting

Figure 2 shows a fixed connector, with male contacts, mounting with thread M12 × 1 and a square flange 20 mm, front mounting.

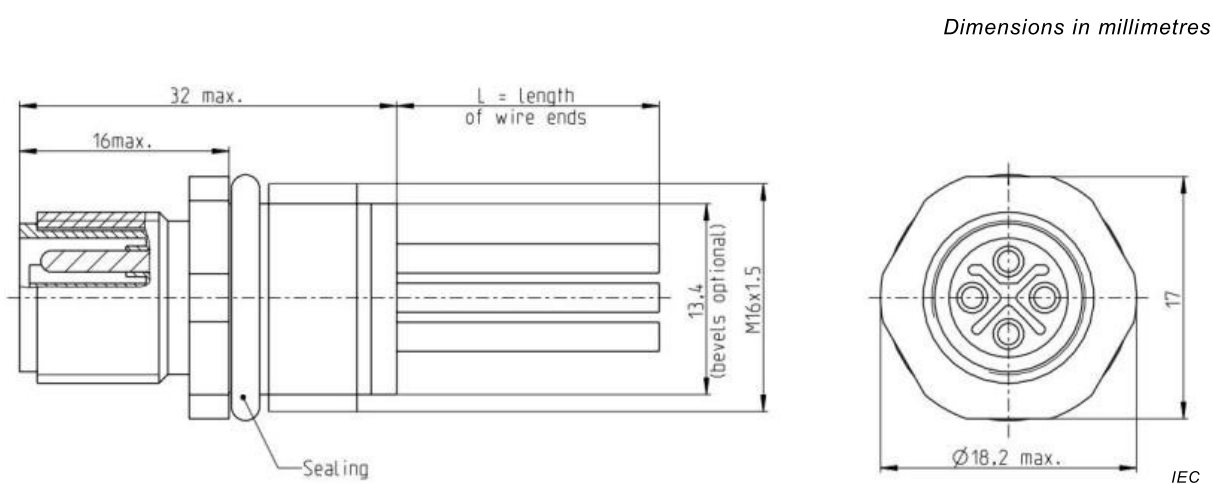


NOTE Orientation of coding, see Annex A.

Figure 2 – Fixed connector, male contacts, mounting with thread M12 x 1, square flange 25 mm, front mounting

5.2.4.3 Style EM

Figure 3 shows a fixed connector, with male contacts, mounting with thread M12 × 1, with wire ends and a single hole mounting thread M16 × 1,5.



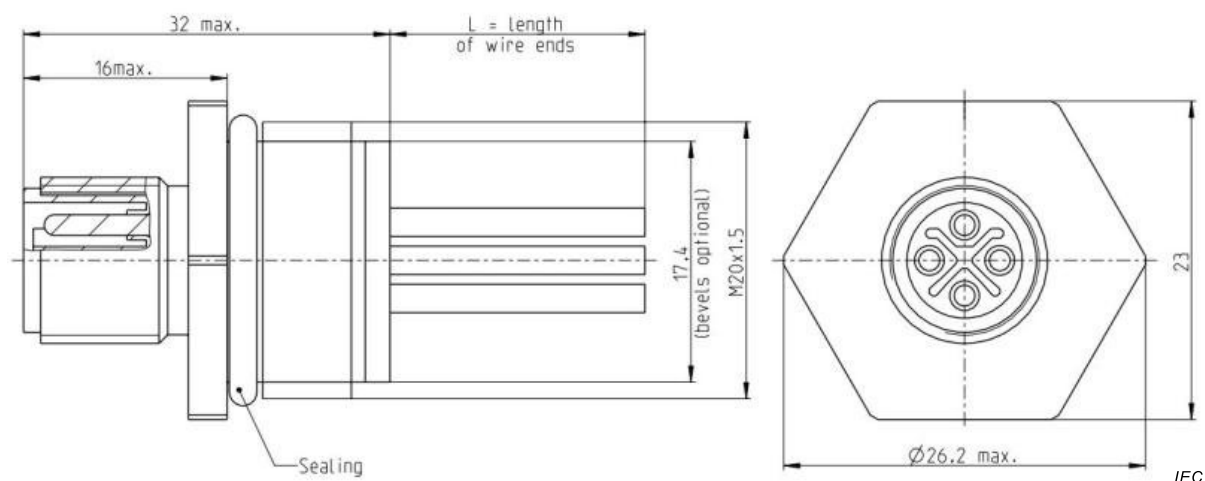
NOTE Orientation of coding, see Annex A.

Figure 3 – Fixed connector, male contacts, mounting with thread M12 x 1, with wire ends, single hole mounting thread M16 x 1,5

5.2.4.4 Style FM

Figure 4 shows fixed connectors, with male contacts, mounting with thread M12 × 1, with wire ends and a single hole mounting thread M20 × 1,5.

Dimensions in millimetres



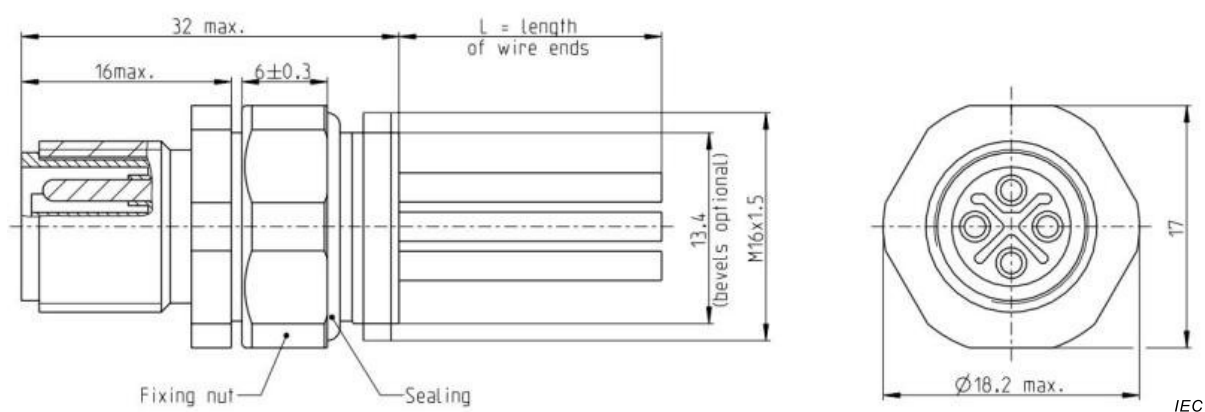
NOTE Orientation of coding, see Annex A.

Figure 4 – Fixed connector, male contacts, mounting with thread M12 x 1, with wire ends, single hole mounting thread M20 x 1,5

5.2.4.5 Style GM

Figure 5 shows fixed connector, male contacts, mounting with thread M12 × 1 with wire ends, a single hole mounting thread M16 × 1,5 and a mounting orientation.

Dimensions in millimetres



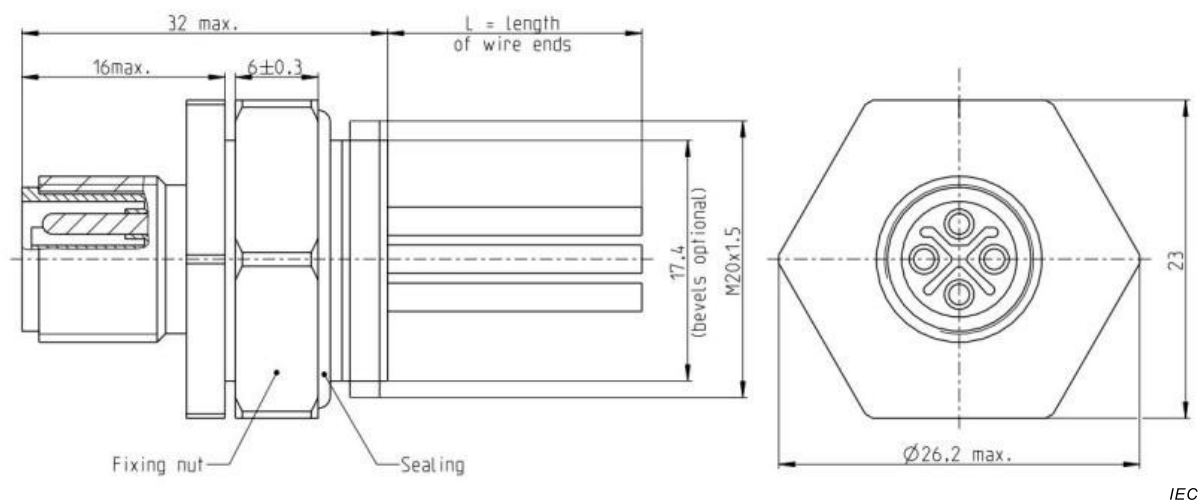
NOTE Orientation of coding, see Annex A.

Figure 5 – Fixed connector, male contacts, mounting with thread M12 × 1 with wire ends, single hole mounting thread M16 × 1,5, mounting orientation

5.2.4.6 Style HM

Figure 6 shows a fixed connector, with male contacts, a mounting with thread M12 × 1, with wire ends, a single hole mounting thread M20 × 1,5 and a mounting orientation.

Dimensions in millimetres



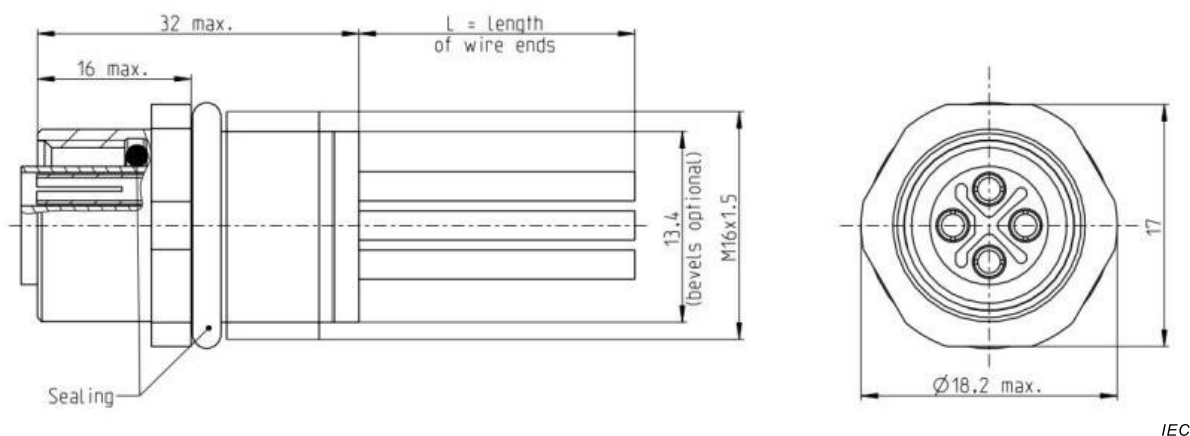
NOTE Orientation of coding, see Annex A.

Figure 6 – Fixed connector, male contacts, mounting with thread M12 × 1, with wire ends, single hole mounting thread M20 × 1,5, mounting orientation

5.2.4.7 Style EF

Figure 7 shows a fixed connector, female contacts, a mounting with thread M12 × 1, with wire ends and a single hole mounting thread M16 × 1,5.

Dimensions in millimetres

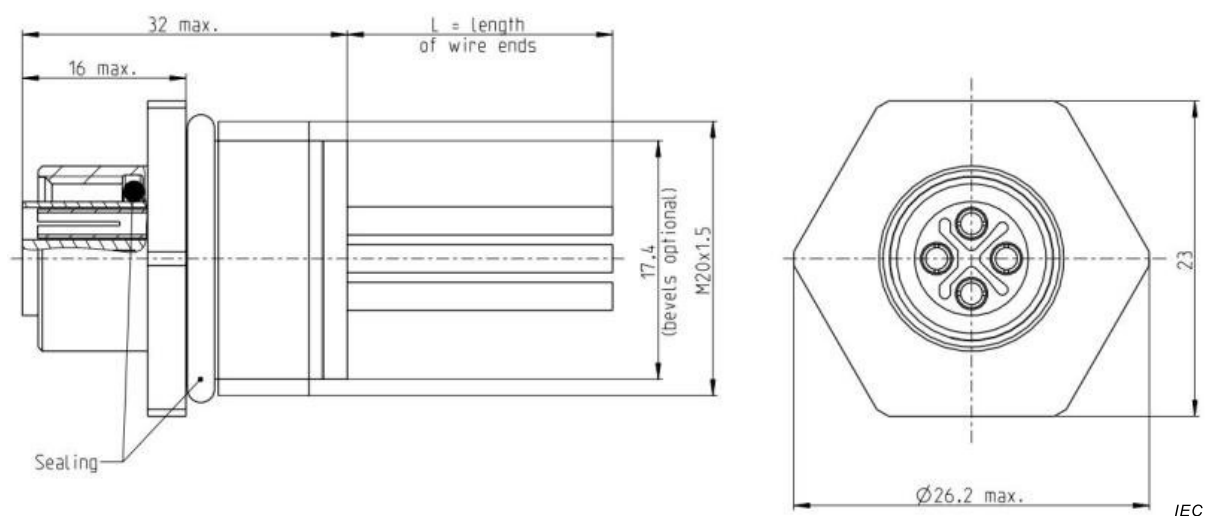


NOTE Orientation of coding, see Annex A.

Figure 7 – Fixed connector, female contacts, mounting with thread M12 × 1, with wire ends, single hole mounting thread M16 × 1,5

5.2.4.8 Style FF

Figure 8 shows a fixed connector, with female contacts, a mounting with thread M12 × 1, with wire ends and a single hole mounting thread M20 × 1,5.

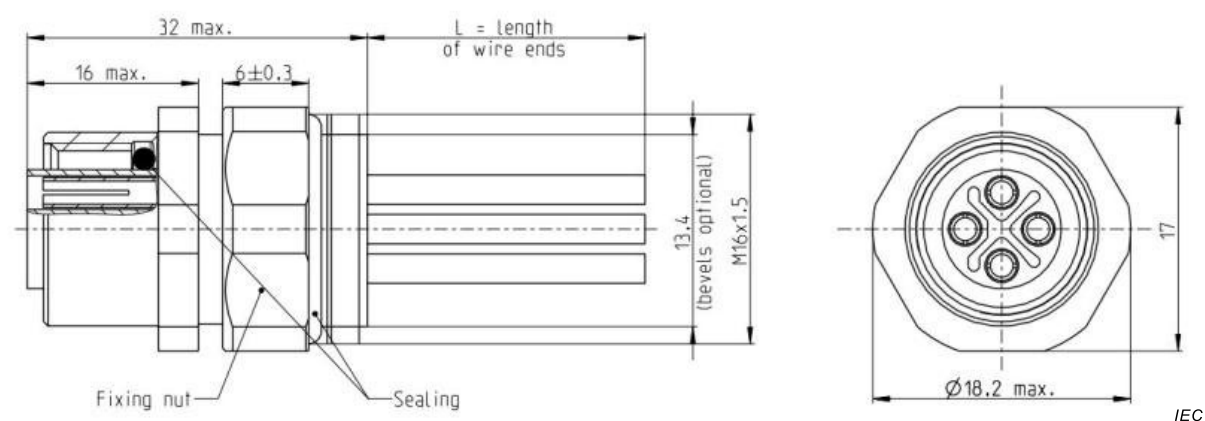
Dimensions in millimetres

NOTE Orientation of coding, see Annex A.

Figure 8 – Fixed connector, female contacts, mounting with thread M12 × 1, with wire ends, single hole mounting thread M20 × 1,5

5.2.4.9 Style GF

Figure 9 shows a fixed connector, with female contacts, a mounting with thread M12 × 1, with wire ends, a single hole mounting thread M16 × 1,5 and a mounting orientation.

Dimensions in millimetres

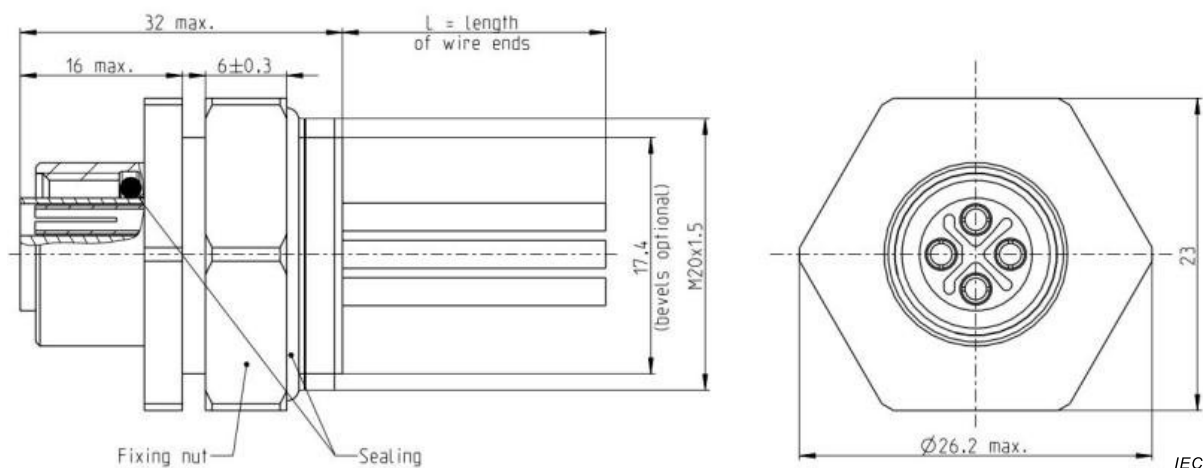
NOTE Orientation of coding, see Annex A.

Figure 9 – Fixed connector, female contacts, mounting with thread M12 × 1, with wire ends, single hole mounting thread M16 × 1,5

5.2.4.10 Style HF

Figure 10 shows a fixed connector, with female contacts, mounting with thread M12 × 1, with wire ends, a single hole mounting thread M20 × 1,5 and a mounting orientation.

Dimensions in millimetres



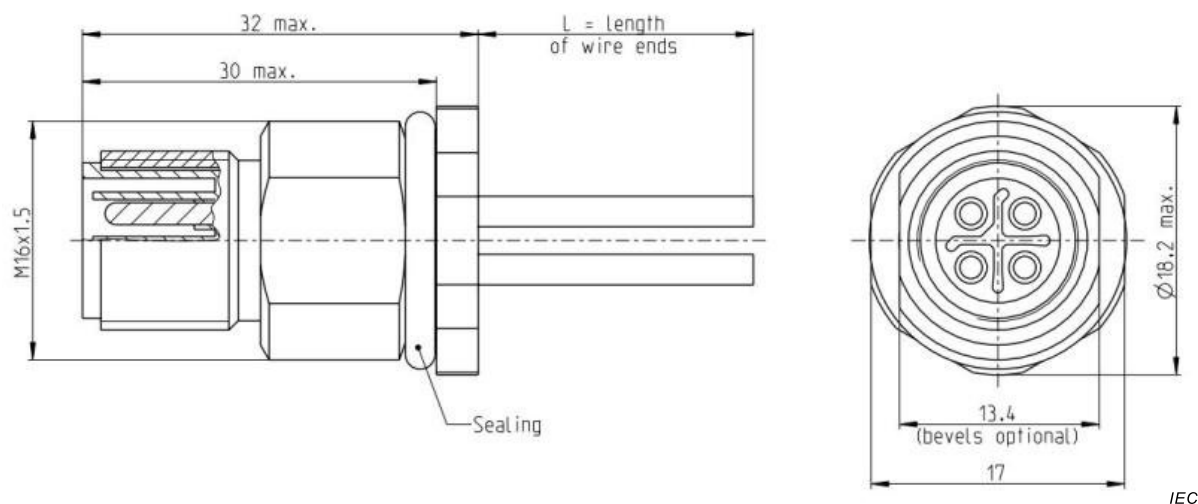
NOTE Orientation of coding, see Annex A.

Figure 10 – Fixed connector, female contacts, mounting with thread M12 × 1, with wire ends, single hole mounting thread M20 × 1,5, mounting orientation

5.2.4.11 Style IM

Figure 11 shows a fixed connector, with male contacts, mounting with thread M12 × 1, with wire ends, a single hole mounting thread M16 x 1,5 rear mounting.

Dimensions in millimetres



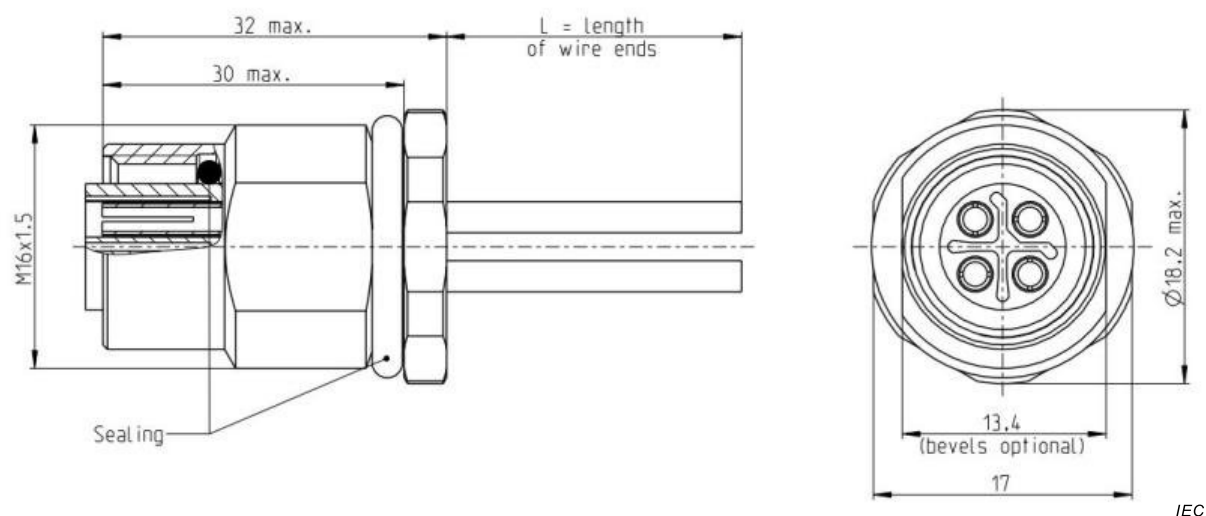
NOTE Orientation of coding, see Annex A.

Figure 11 – Fixed connector, male contacts, mounting with thread M12 × 1, with wire ends, single hole mounting thread M16 × 1,5, mounting orientation

5.2.4.12 Style IF

Figure 12 shows a fixed connector, with female contacts, mounting with thread M12 x 1, with wire ends, a single hole mounting thread M 16 x 1,5, rear mounting.

Dimensions in millimetres



IEC

NOTE Orientation of coding, see Annex A.

Figure 12 – Fixed connector, female contacts, mounting with thread M12 × 1, with wire ends, single hole mounting thread M16 × 1,5, mounting orientation

5.2.5 Free connectors

5.2.5.1 General

Table 4 shows styles of free connectors.

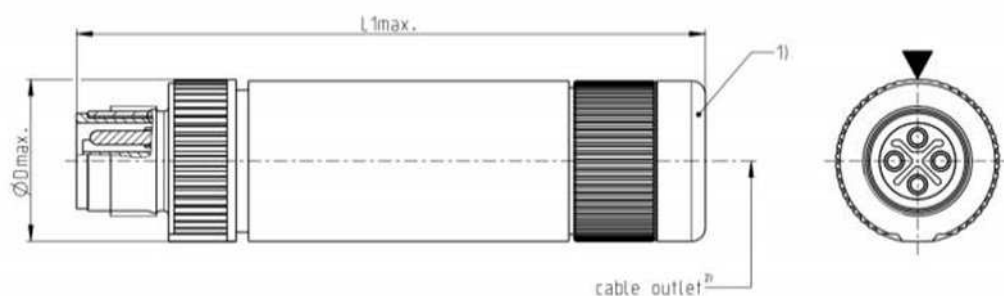
Table 4 – Styles of free connectors

Style	Description
JM	Rewireable connector, male contacts, straight version, with locking nut ^a
KM	Rewireable connector, male contacts, right angled version, with locking nut ^a
LM	Non-rewireable connector, male contacts, straight version, with locking nut ^a
MM	Non-rewireable connector, male contacts, right angled version, with locking nut
JF	Rewireable connector, female contacts, straight version, with locking nut
KF	Rewireable connector, female contacts, right angled version, with locking nut
LF	Non-rewireable connector, female contacts, straight version, with locking nut
MF	Non-rewireable connector, female contacts, right angled version, with locking nut
^a Knurled ring or hexagonal ring upon agreement.	

5.2.5.2 Style JM

Figure 13 shows a rewireable connector, with male contacts, a straight version with locking nut.

Dimensions in millimetres



IEC

NOTE 1 Cable outlet alternatively inside.

NOTE 2 Cable outlet diameter range upon agreement.

NOTE 3 ▼ Orientation of coding, see Annex A.

Figure 13 – Rewirable connector, male contacts, straight version, with locking nut

Table 5 shows the dimensions of style JM.

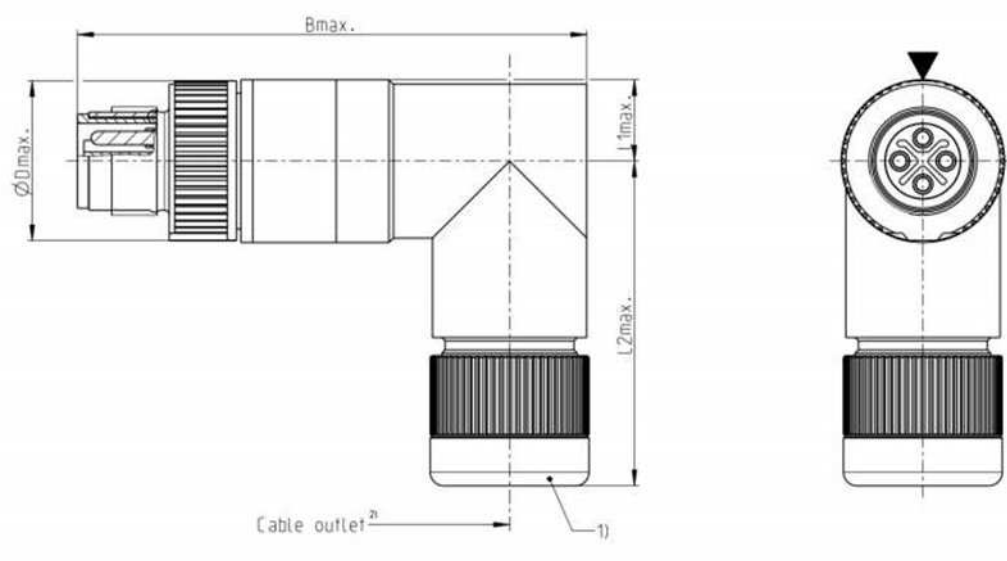
Table 5 – Dimensions of style JM, figure 13

Dimensions in millimetres

Coding	ØD max.	L1
S, T	23	70
E, F, K, L, M	25	70

5.2.5.3 Style KM

Figure 14 shows a rewirable connector with male contacts, right angled version with locking nut.

Dimensions in millimetres

IEC

NOTE 1 Cable outlet alternatively inside.

NOTE 2 Cable outlet diameter range upon agreement.

NOTE 3 ▼ Orientation of coding, see Annex A.

Figure 14 – Rewireable connector, male contacts, right angled version, with locking nut

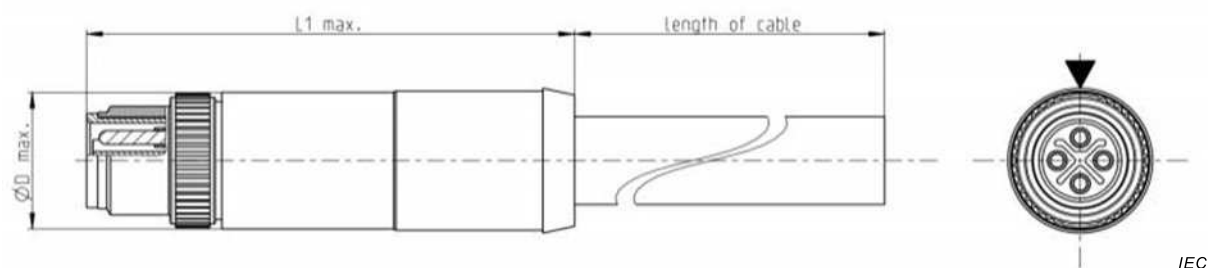
Table 6 shows the dimensions of style KM.

Table 6 – Dimensions of style KM, figure 14*Dimensions in millimetres*

Coding	ØD max.	ØB max.	L1 max.	L2 max.
S, T	23	50	13	40
E, F, K, L, M	25	50	14	40

5.2.5.4 Style LM

Figure 15 shows a non-rewireable connector, with male contacts and a straight version with locking nut.

Dimensions in millimetres

IEC

NOTE1 ▼ Orientation of coding, see Annex A.

Figure 15 – Non-rewireable connector, male contacts, straight version, with locking nut

Table 7 shows the dimensions of style LM.

Table 7 – Dimensions of style LM, figure 15

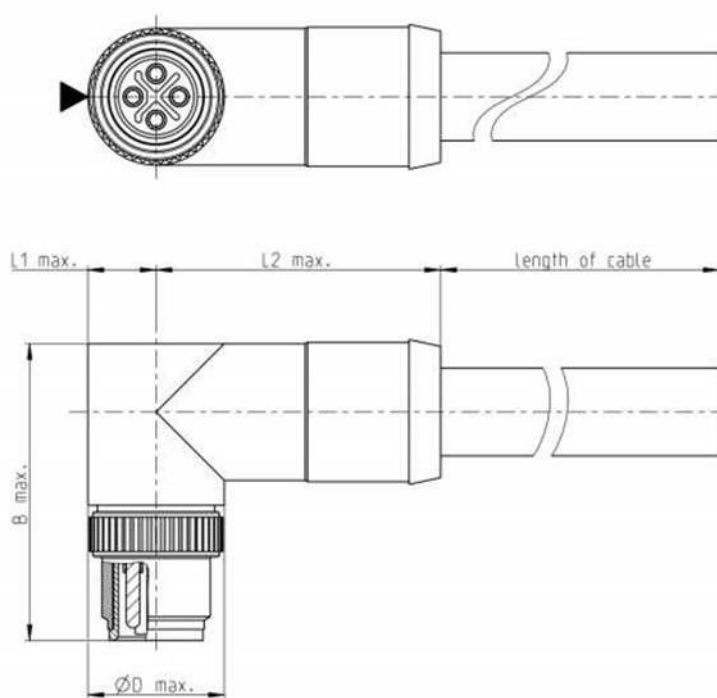
Dimensions in millimetres

Coding	ØD max.	L1
S, T	16	57
E, F, K, L, M	18	65

5.2.5.5 Style MM

Figure 16 shows a non-rewireable connector, with male contacts, a right angled version with locking nut.

Dimensions in millimetres



IEC

NOTE ▼ Orientation of coding, see Annex A.

Figure 16 – Non-rewireable connector, male contacts, right angled version, with locking nut

Table 8 shows the dimensions of style MM.

Table 8 – Dimensions of style MM, figure 16

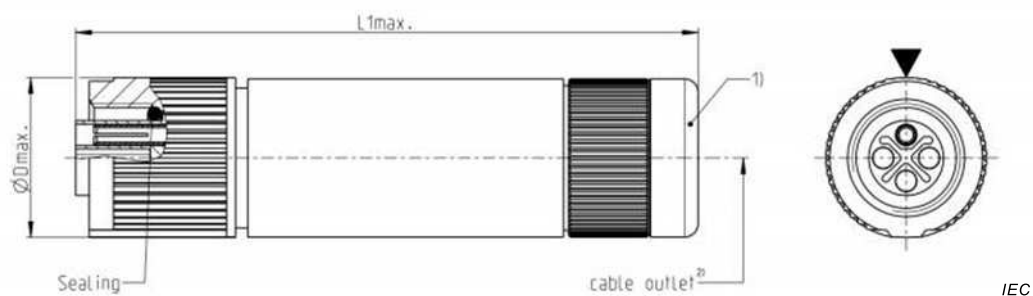
Dimensions in millimetres

Coding	ØD max.	B max.	L1 max.	L2 max.
S, T	16	36	8	33,5
E, F, K, L, M	18	41	9	40

5.2.5.6 Style JF

Figure 17 shows a rewireable connector, with female contacts, straight version with locking nut.

Dimensions in millimetres



NOTE 1 Cable outlet alternatively inside.

NOTE 2 Cable outlet diameter range upon agreement

NOTE 3 ▼ Orientation of coding, see Annex A.

Figure 17 – Rewireable connector, female contacts, straight version, with locking nut

Table 9 shows the dimensions of style JF.

Table 9 – Dimensions of style JF, figure 17

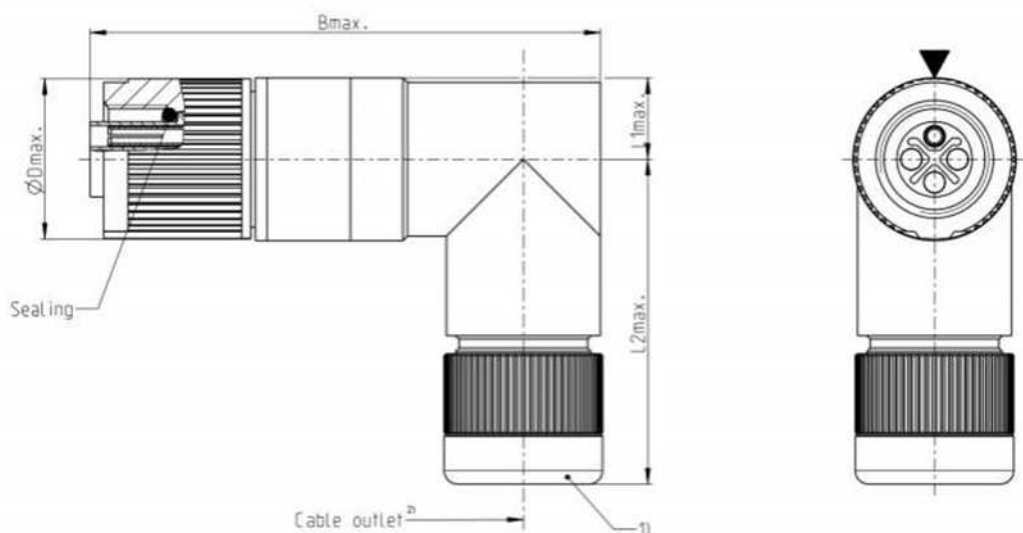
Dimensions in millimetres

Coding	Ø Dmax.	L1
S, T	23	70
E, F, K, L, M	25	70

5.2.5.7 Style KF

Figure 18 shows a rewirable connector, with female contacts and a right angled version with locking nut.

Dimensions in millimetres



IEC

NOTE 1 Cable outlet alternatively inside.

NOTE 2 Cable outlet diameter range upon agreement

NOTE 3 ▼ Orientation of coding, see Annex A.

Figure 18 – Rewirable connector, female contacts, right angled version, with locking nut

Table 10 shows the dimensions of style KF.

Table 10 – Dimensions of style KF, figure 18

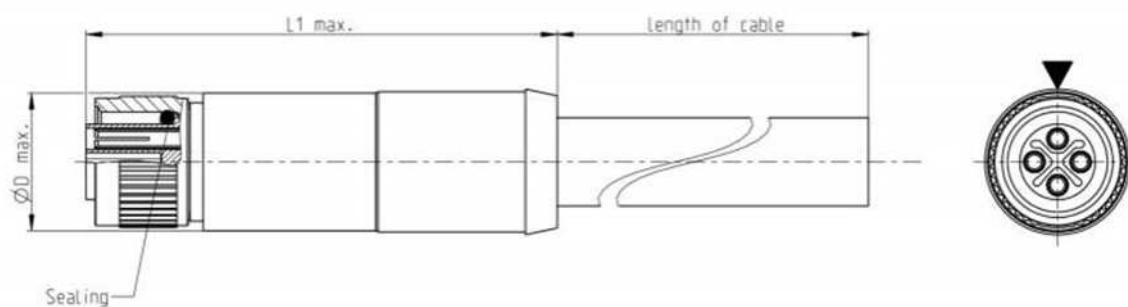
Dimensions in millimetres

Coding	Ø Dmax.	B max.	L1 max.	L2 max.
S, T	23	50	13	40
E, F, K, L, M	25	50	14	40

5.2.5.8 Style LF

Figure 19 shows a non-rewireable connector, with female contacts and a straight version with locking nut.

Dimensions in millimetres



IEC

NOTE ▼ Orientation of coding, see Annex A.

Figure 19 – Non-rewireable connector, female contacts, straight version, with locking nut

Table 11 shows the dimensions of style LF.

Table 11 – Dimensions of style LF, figure 19

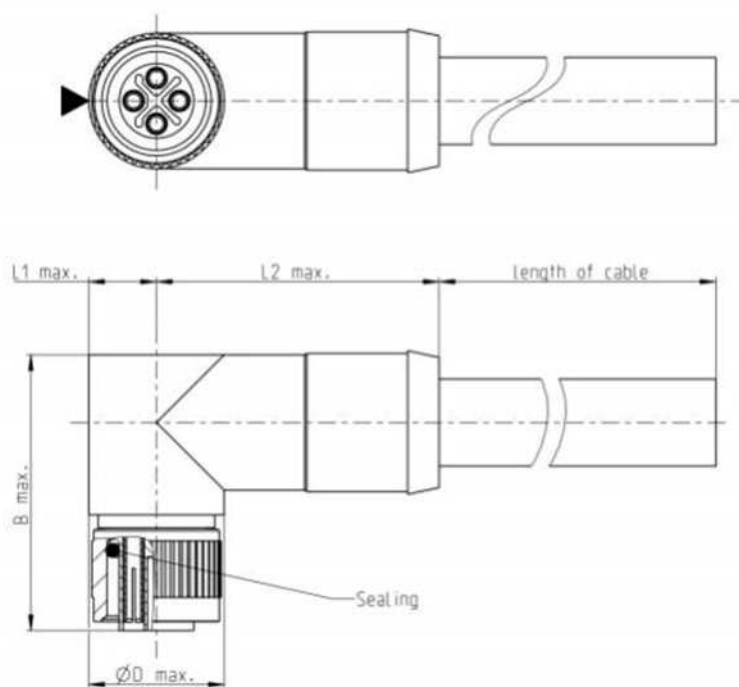
Dimensions in millimetres

Coding	Ø Dmax.	L1
S, T	16	55
E, F, K, L, M	18	63

5.2.5.9 Style MF

Figure 20 shows a non-rewireable connector, with female contacts and a right angled version with locking nut.

Dimensions in millimetres



IEC

NOTE ▼ Orientation of coding, see Annex A.

Figure 20 – Non-rewireable connector, female contacts, right angled version, with locking nut

Table 12 shows the dimensions of style MF.

Table 12 – Dimensions of style MF, figure 20

Dimensions in millimetres

Coding	Ø Dmax.	B max.	L1 max.	L2 max.
S, T	16	35	8	33,5
E, F, K, L, M	18	40	9	40

5.3 Interface dimensions

5.3.1 E-coding

5.3.1.1 Male side

Figure 21 shows the male E-coding.

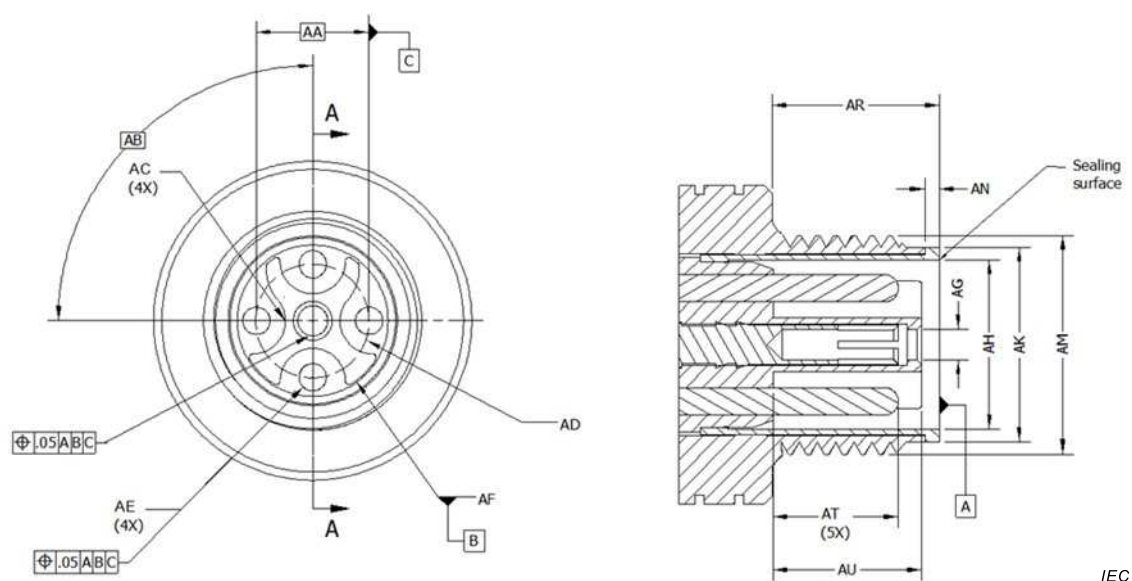
Dimensions in millimetres**Figure 21 – Male side E-coding**

Table 13 shows the dimensions for Figure 21.

Table 13 – Dimensions for figure 21*Dimensions in millimetres*

Letter	Minimum	Nominal	Maximum
AA		Ø 6,20	
AB	89,5°	90,0°	90,5°
AC	Ø 3,55	Ø 3,60	Ø 3,65
AD	Ø 5,05	Ø 5,1	Ø 5,15
AE	Ø 1,47	Ø 1,50	Ø 1,53
AF	Ø 8,05	Ø 8,10	Ø 8,15
AG	Ø 1,60	Ø 1,65	Ø 1,70
AH	Ø 9,3	Ø 9,3	Ø 9,35
AK	Ø 10,64	Ø 10,69	Ø 10,74
AM		M12x1 thread	
AN	0,70	0,80	0,90
AR	10,0		
AT	7,25	7,40	7,55
AU	8,60	8,70	8,80
AV	1,35	1,40	1,45
AW	Ø 2,15	Ø 2,20	Ø 2,25

5.3.1.2 Female side

Figure 22 shows the female E-coding.

Dimensions in millimetres

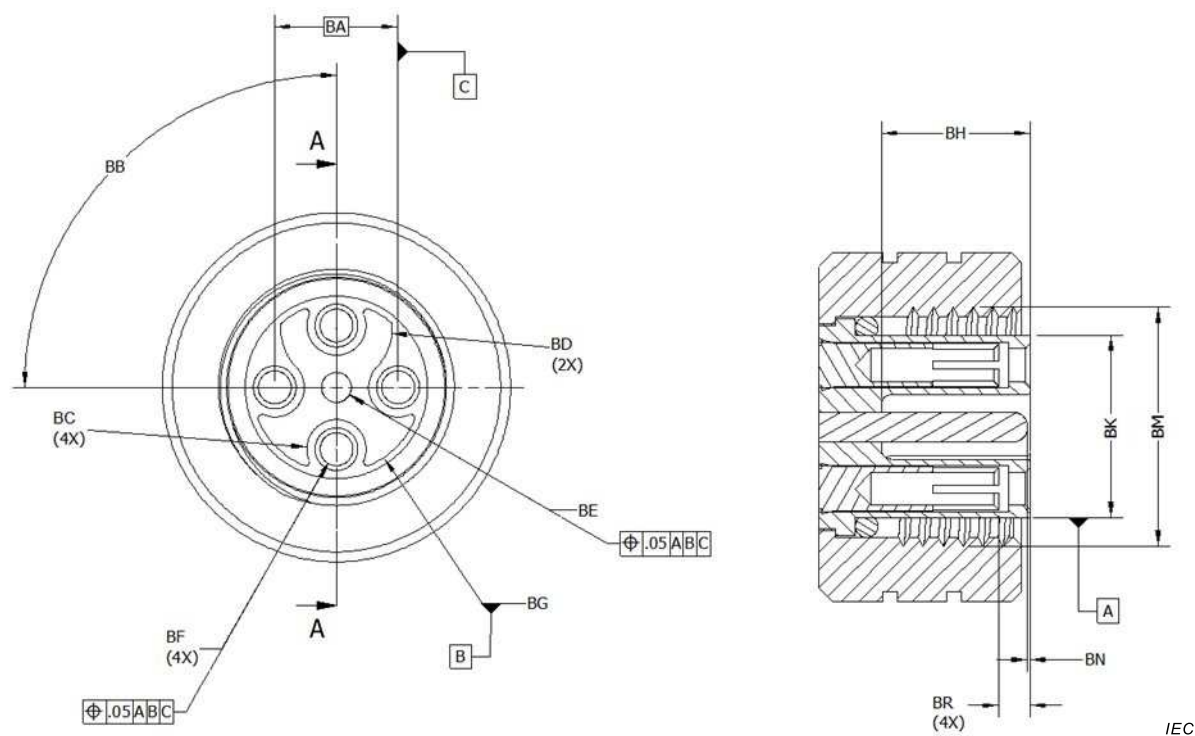


Figure 22 – Female side E-coding

Table 14 shows the dimensions for Figure 22.

Table 14 – Dimensions for Figure 22

Dimensions in millimetres

Letter	Minimum	Nominal	Maximum
BA		Ø 6,20	
BB	89,5°	90,0°	90,5°
BC	Ø 2,92	Ø 2,97	Ø 3,03
BD	Ø 5,55	Ø 5,6	Ø 5,65
BE	Ø 1,47	Ø 1,5	Ø 1,53
BF	Ø 1,6	Ø 1,65	Ø 1,7
BG	Ø 8,45	Ø 8,5	Ø 8,55
BH	7,45	7,5	7,55
BK	Ø 9,1	Ø 9,15	Ø 9,2
BM		M12x1 thread	
BN	0,1	0,15	0,2
BR	1,55	1,6	1,6

5.3.1.3 Contact position E-coding, front view

Figure 23 shows the contact position for E-coding front view.

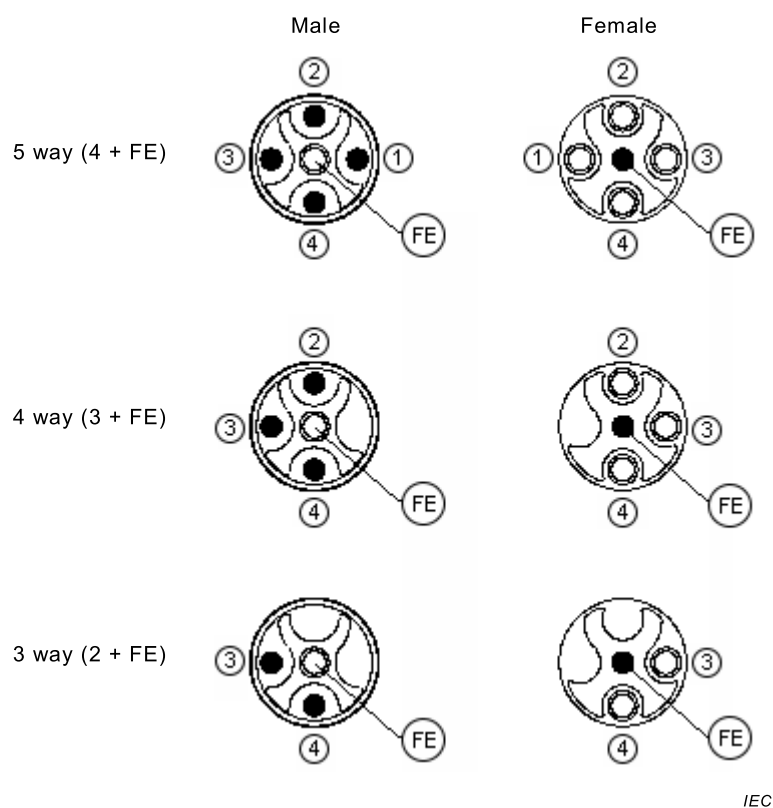


Figure 23 – Contact position for E-coding front view

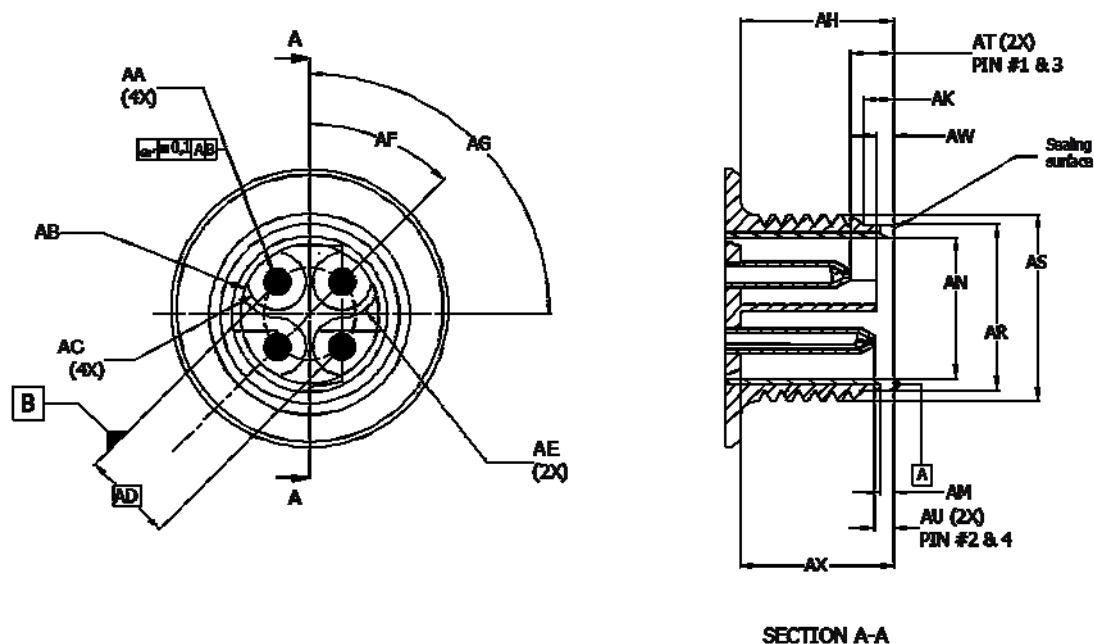
The contact marking shall be on the termination side of the connector insert as long as the size of the component allows the placement there.

5.3.2 F-coding

5.3.2.1 Male side

Figure 24 shows the F-coding male side.

Dimensions in millimetres



IEC

Figure 24 – Male side F-coding

Table 15 shows the dimensions for Figure 24.

Table 15 – Dimensions for Figure 24

Dimensions in millimetres

Letter	Minimum	Nominal	Maximum
AA	Ø 1,47	Ø 1,50	
AB	Ø 8,30	Ø 8,30	Ø 8,40
AC	Ø 3,42	Ø 3,42	Ø 3,52
AD		Ø 5,50	
AE	Ø 4,36	Ø 4,36	Ø 4,46
AF	44,5°	45,0°	45,5°
AG	89,5°	90,0°	90,5°
AH	10,0		
AK			2,00
AM	0,70	0,80	0,90
AN	Ø 9,30	Ø 9,30	Ø 9,33
AR	Ø 10,60	Ø 10,70	Ø 10,80
AS		M12x1 thread	
AT	2,50	2,60	2,65
AU	1,00	1,10	1,20
AW	0,93	0,95	0,97
AX	9,10	9,20	9,20

5.3.2.2 Female side

Figure 25 shows the female side F-coding.

Dimensions in millimetres

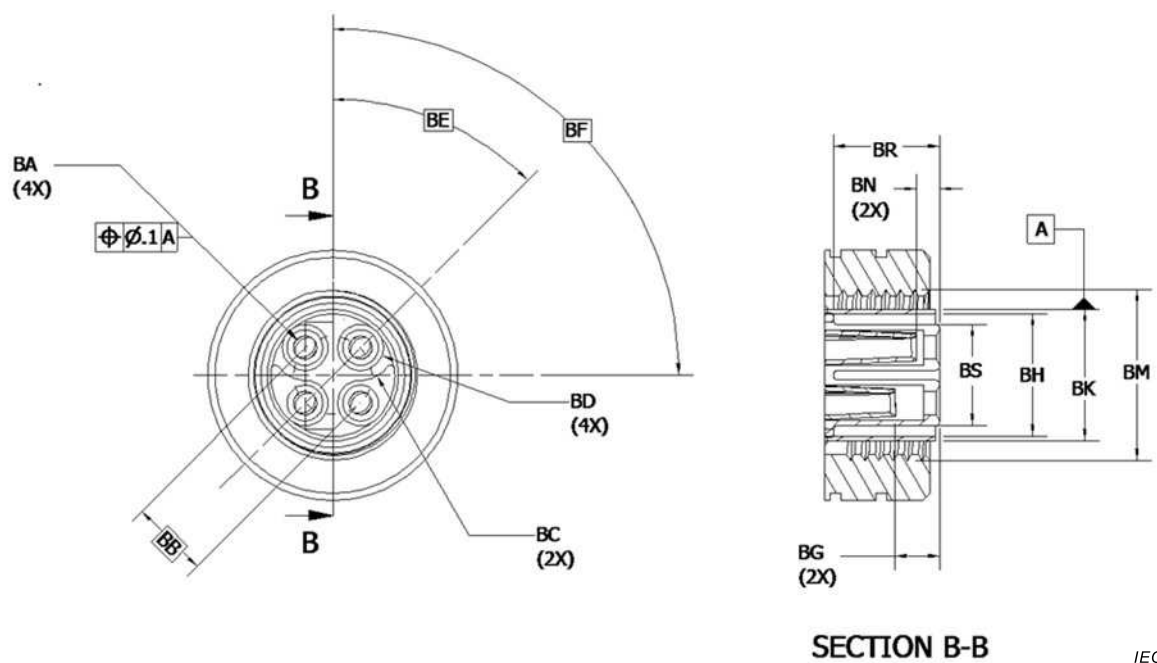


Figure 25 – Female side F-coding

Table 16 shows the dimensions for Figure 25.

Table 16 – Dimensions for Figure 25

Dimensions in millimetres

Letter	Minimum	Nominal	Maximum
BA	Ø 1,60	Ø 1,63	Ø 1,66
BB		Ø 5,50	
BC	Ø 4,53	Ø 4,58	Ø 4,58
BD	Ø 3,15	Ø 3,20	Ø 3,20
BE	44,5°	45,0°	45,5°
BF	89,5°	90,0°	90,5°
BG	3,01	3,11	3,21
BH	Ø 8,55	Ø 8,60	Ø 8,60
BK	Ø 9,20	Ø 9,20	Ø 9,25
BM		M12x1 thread	
BN	1,52	1,62	1,72
BR	7,40	7,45	7,50
BS	Ø 7,05	Ø 7,10	Ø 7,15

5.3.2.3 Contact position F-coding, front view

Figure 26 shows the contact position for F-coding front view.

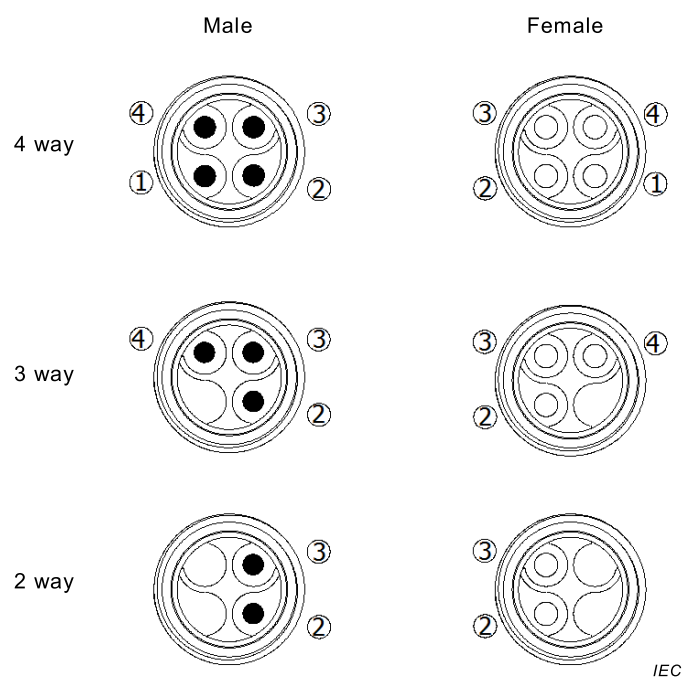


Figure 26 – Contact position for F-coding front view

The contact marking shall be on the termination side of the connector insert as long as the size of the component allows the placement there.

5.3.3 K-coding

5.3.3.1 Male side

Figure 27 shows the male side K-coding.

Dimensions in millimetres

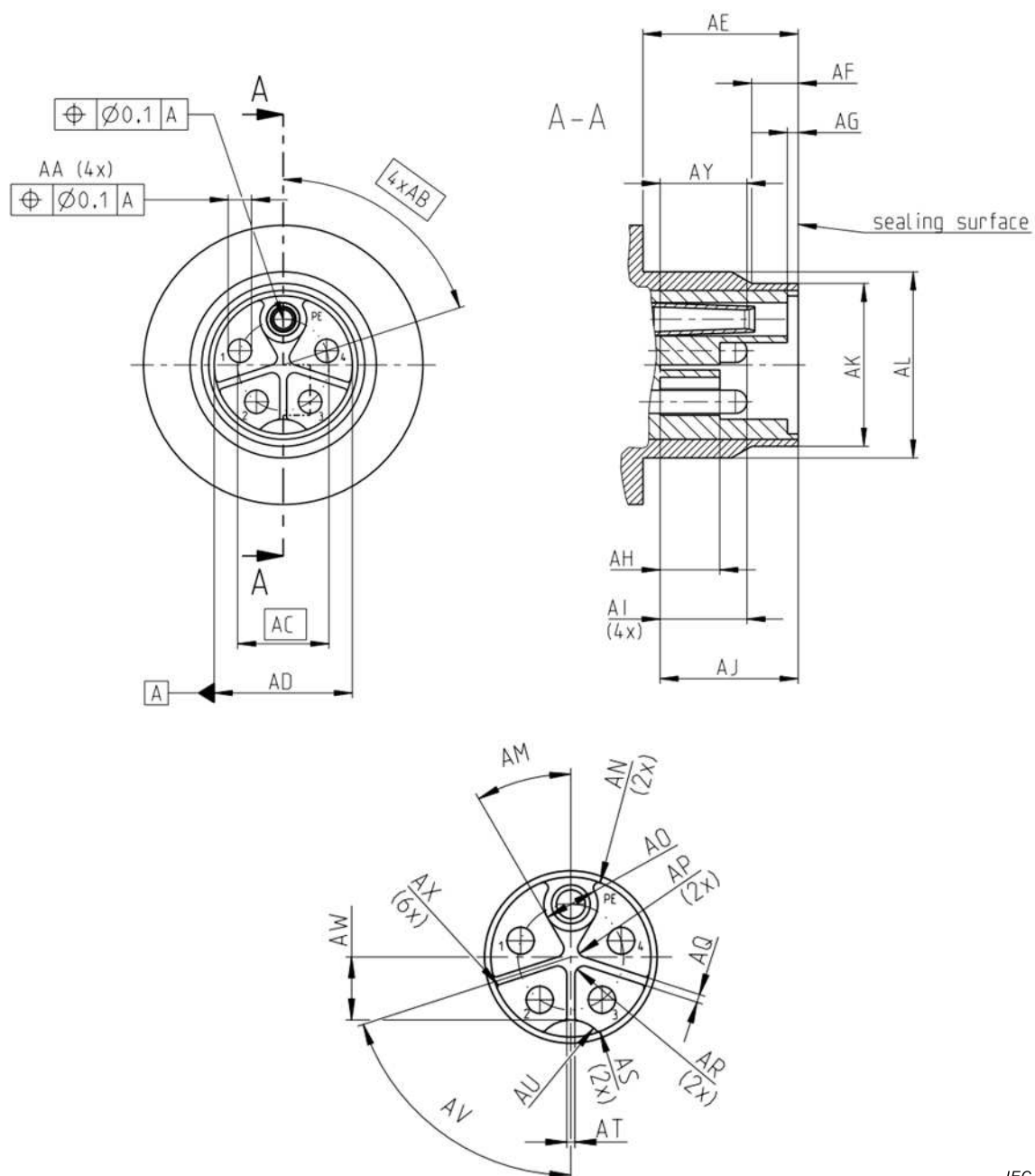


Figure 27 – K-coding male side

IEC

Table 17 shows the dimensions for Figure 27.

Table 17 – Dimensions for Figure 27

Dimensions in millimetres

Letter	Minimum	Nominal	Maximum
AA	Ø 1,47	Ø 1,5	
AB		72°	
AC		Ø 5,88	
AD	Ø 8,8	Ø 8,9	
AE	10		
AF			3
AG		0,7	
AH	3,75	3,85	
AI	5,2	5,6	
AJ	8,7	8,9	
AK	Ø 10,2	Ø 10,5	Ø 10,7
AL		M12x1 thread	
AM	29,5°	30°	30,5°
AN	R 0,25	R 0,3	
AO		Ø 3	Ø 3,1
AP	R 0,45	R 0,5	
AQ	0,35	0,45	
AR	R 0,45	R 0,5	
AS	R 0,15	R 0,2	
AT	0,35	0,45	
AU	R 1,9	R 2	
AV	71,5°	72°	72,5°
AW		3,5	3,6
AX	R 0,15	R 0,2	
AY	5,2		

5.3.3.2 Female side

Figure 28 shows the K-coding female side.

Dimensions in millimetres

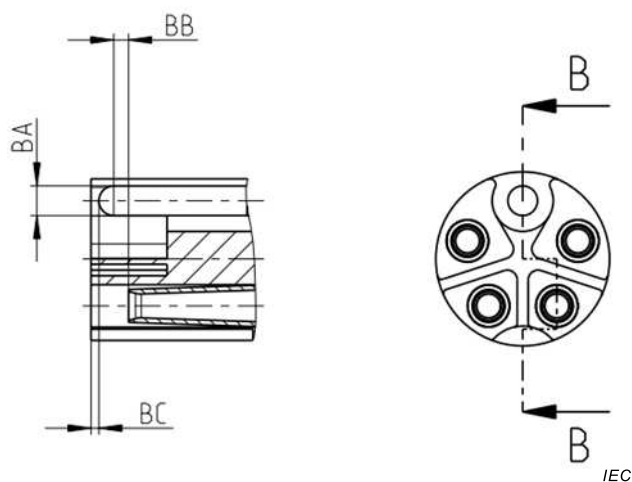


Figure 28 – K-coding female side

Table 18 shows the dimensions for Figure 28.

Table 18 – Dimensions for Figure 28

Dimensions in millimetres

Letter	Minimum	Nominal	Maximum
BA	Ø 1,47	Ø 1,5	
BB	0,9		
BC	0,0		

5.3.3.3 Contact position K-coding, front view

Figure 29 shows the contact position K-coding front view.

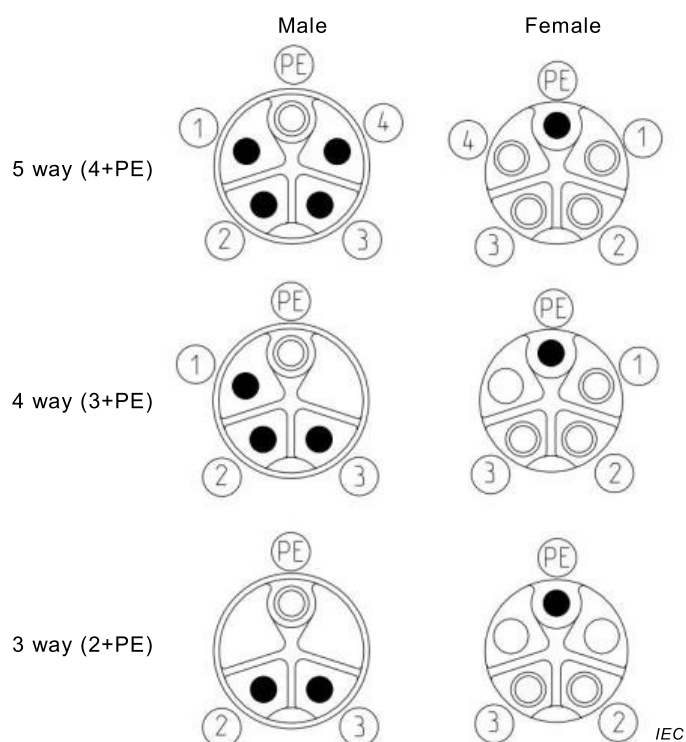


Figure 29 – Contact position K-coding front view

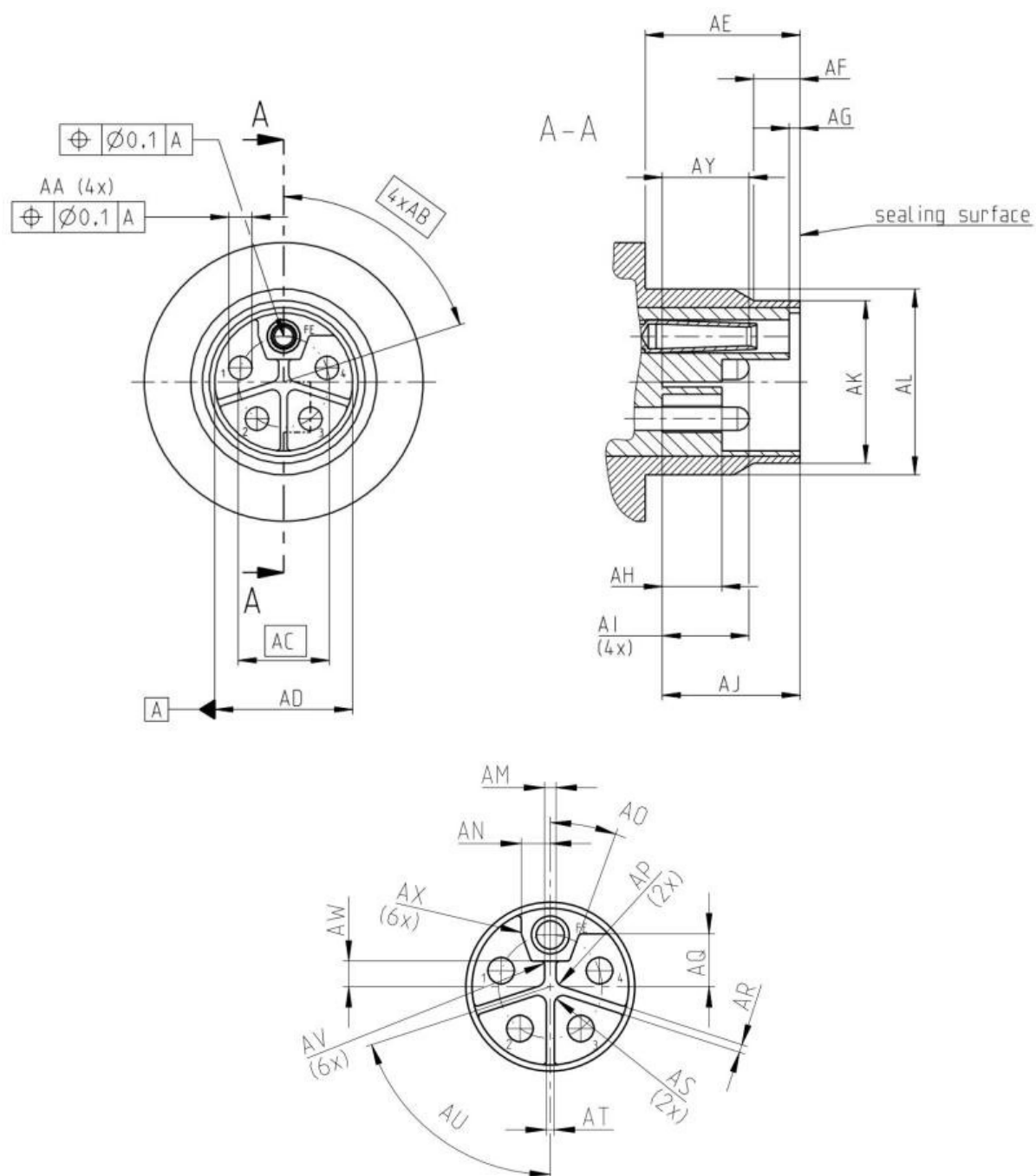
The contact marking shall be on the termination side of the connector insert as long as the size of the component allows the placement there.

5.3.4 L-coding

5.3.4.1 Male side

Figure 30 shows the male L-coding with one female contact.

Dimensions in millimetres



IEC

Figure 30 – L-coding male side with one female contact

Table 19 shows the dimensions for Figure 30.

Table 19 – Dimensions for Figure 30

Dimensions in millimetres

Letter	Minimum	Nominal	Maximum
AA	Ø 1,47	Ø 1,5	
AB		72°	
AC		Ø 5,88	
AD	Ø 8,8	Ø 8,9	
AE	10		
AF			3,0
AG		0,7	0,9
AH	3,75	3,85	
AI	5,2	5,6	
AJ	8,7	8,9	
AK	Ø 10,2	Ø 10,5	Ø 10,7
AL		M12x1 thread	
AM	0,55	0,65	
AN	1,55	1,65	
AO		20°	21°
AP	R 0,45	R 0,5	
AQ		3	3,1
AR	0,35	0,45	
AS	R 0,45	R 0,5	
AT	0,35	0,45	
AU	71,5°	72°	72,5°
AV	R 0,15	R 0,2	
AW		1,45	1,55
AX	R 0,15	R 0,2	
AY	5,2		

5.3.4.2 Female side

Figure 31 shows the female L-coding with one male contact.

Dimensions in millimetres

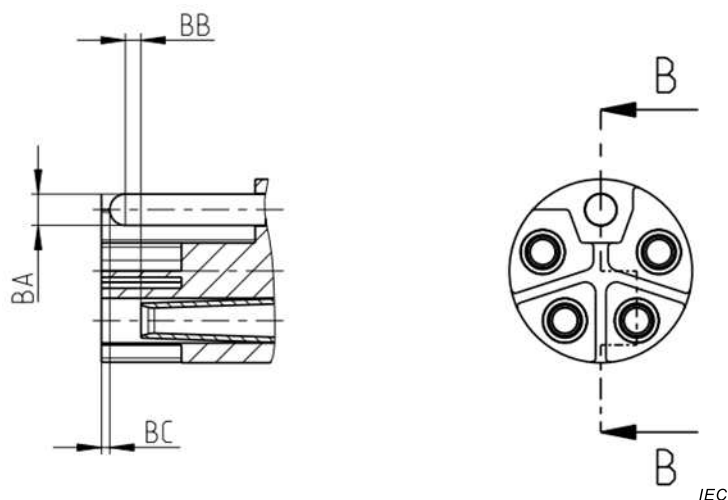


Figure 31 – L-coding female side with one male contact

Table 20 shows the dimensions for Figure 31.

Table 20 – Dimensions for Figure 31

Dimensions in millimetres

Letter	Minimum	Nominal	Maximum
BA	Ø 1,47	Ø 1,5	
BB	0,9		
BC	0,0		

5.3.4.3 Contact position L-coding, front view

Figure 32 shows the contact position L-coding front view.

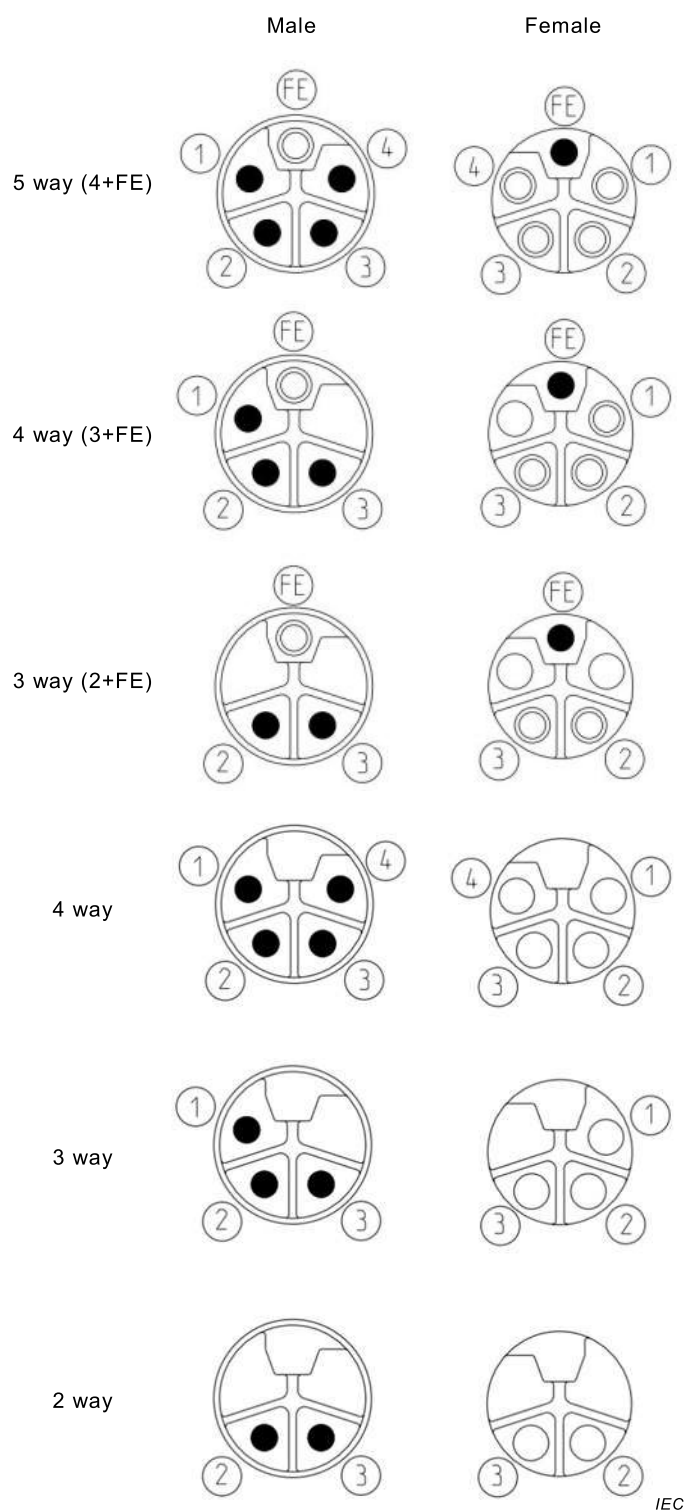


Figure 32 – Contact position L-coding front view

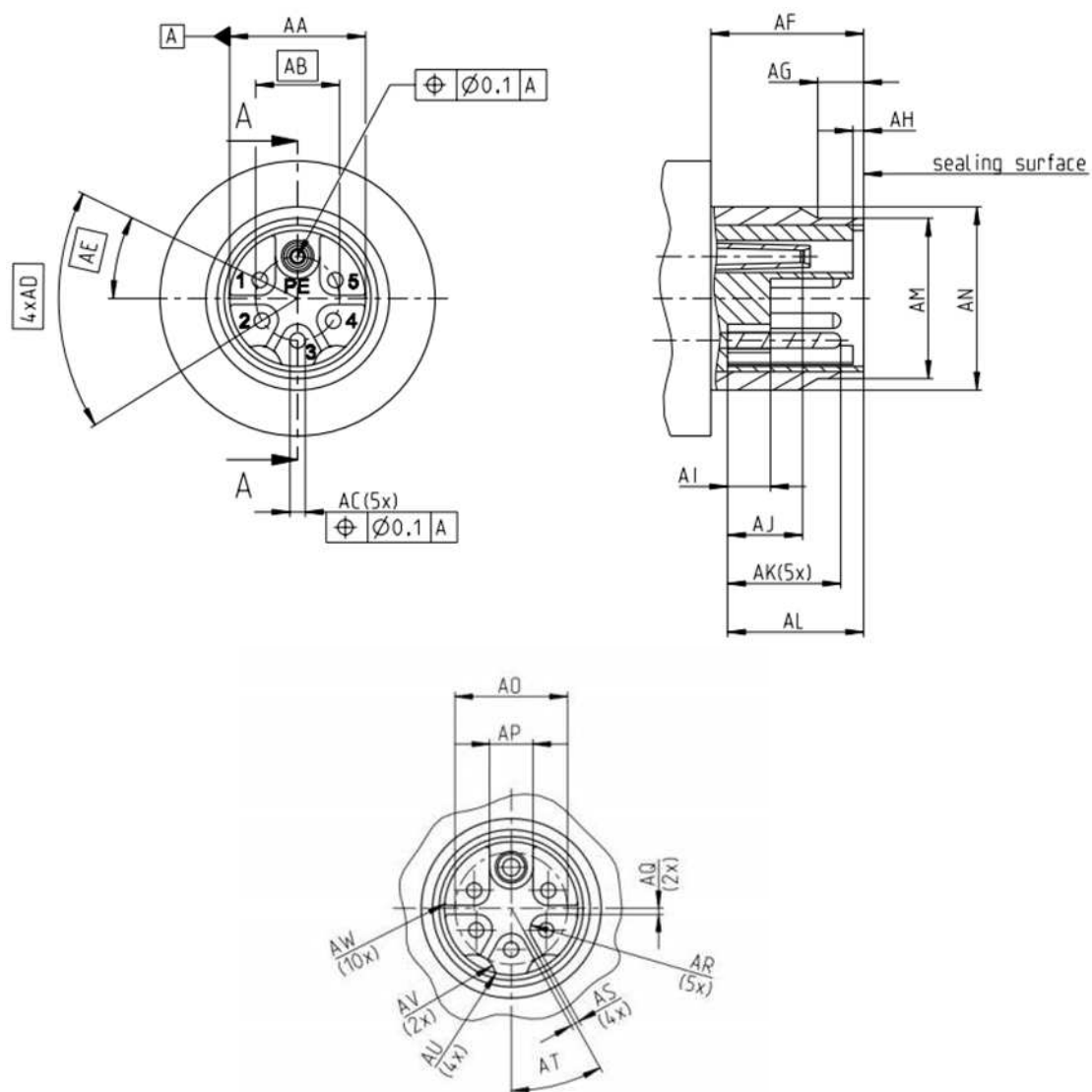
The contact marking shall be on the termination side of the connector insert as long as the size of the component allows the placement there.

5.3.5 M-coding

5.3.5.1 Male side

Figure 33 shows the male M-coding with PE female.

Dimensions in millimetres



IEC

Figure 33 – M-coding male site

Table 21 shows the dimensions for Figure 33.

Table 21 – Dimensions for Figure 33

Dimensions in millimetres

Letter	Minimum	Nominal	Maximum
AA	Ø 8,8	Ø 8,9	
AB		Ø 5,5	
AC	Ø 0,97	Ø 1	Ø 1,03
AD		58°	
AE		26°	
AF	10,0		
AG			3,0
AH		0,7	0,9
AI	2,7	2,8	
AJ	5,1		
AK	6,7	6,9	7,1
AL	8,7	8,9	
AM	Ø 10,2	Ø 10,5	Ø 10,7
AN		M12x1 thread	
AO		Ø 7,5	Ø 7,7
AP	2,80	2,9	
AQ	0,35	0,4	
AR	R 0,9	R 1	
AS	0,5	0,6	
AT	28,5°	29°	29,5°
AU	R 0,15	R 0,2	
AV	R 1,4	R 1,5	
AW			R 0,2

5.3.5.2 Female side

Figure 34 shows the M-coding female side.

Dimensions in millimetres

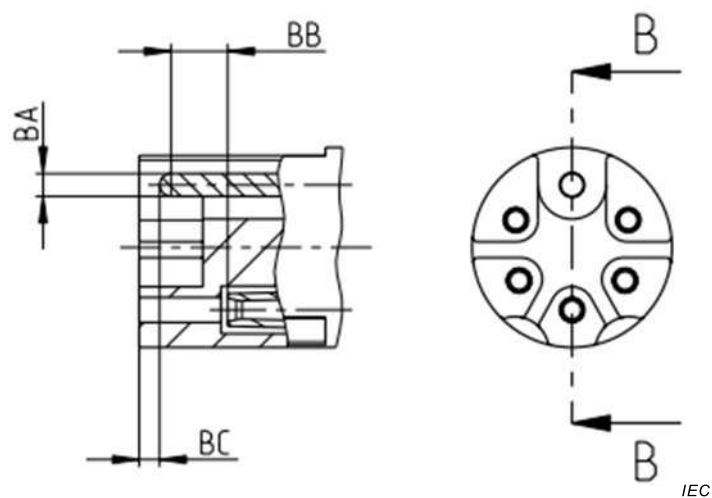


Figure 34 – M-coding female side

Table 22 shows the dimensions for Figure 34.

Table 22 – Dimensions for Figure 34

Dimensions in millimetres

Letter	Minimum	Nominal	Maximum
BA	Ø 0,97	Ø 1	Ø 1,03
BB	2,5		
BC	0,3		

5.3.5.3 Contact position M-coding, front view

Figure 35 shows the contact position M-coding front view.

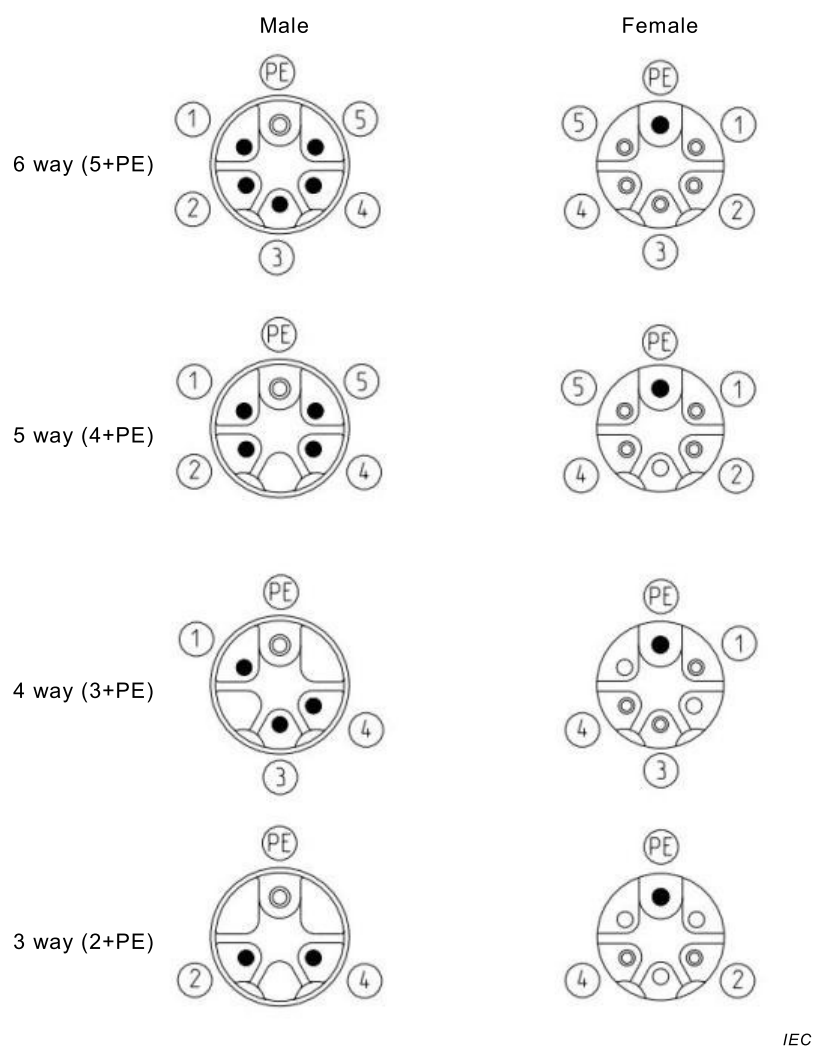


Figure 35 – Contact position M-coding front view

The contact marking shall be on the termination side of the connector insert as long as the size of the component allows the placement there.

5.3.6 S-coding

5.3.6.1 Male side

Figure 36 shows the S-coding male side.

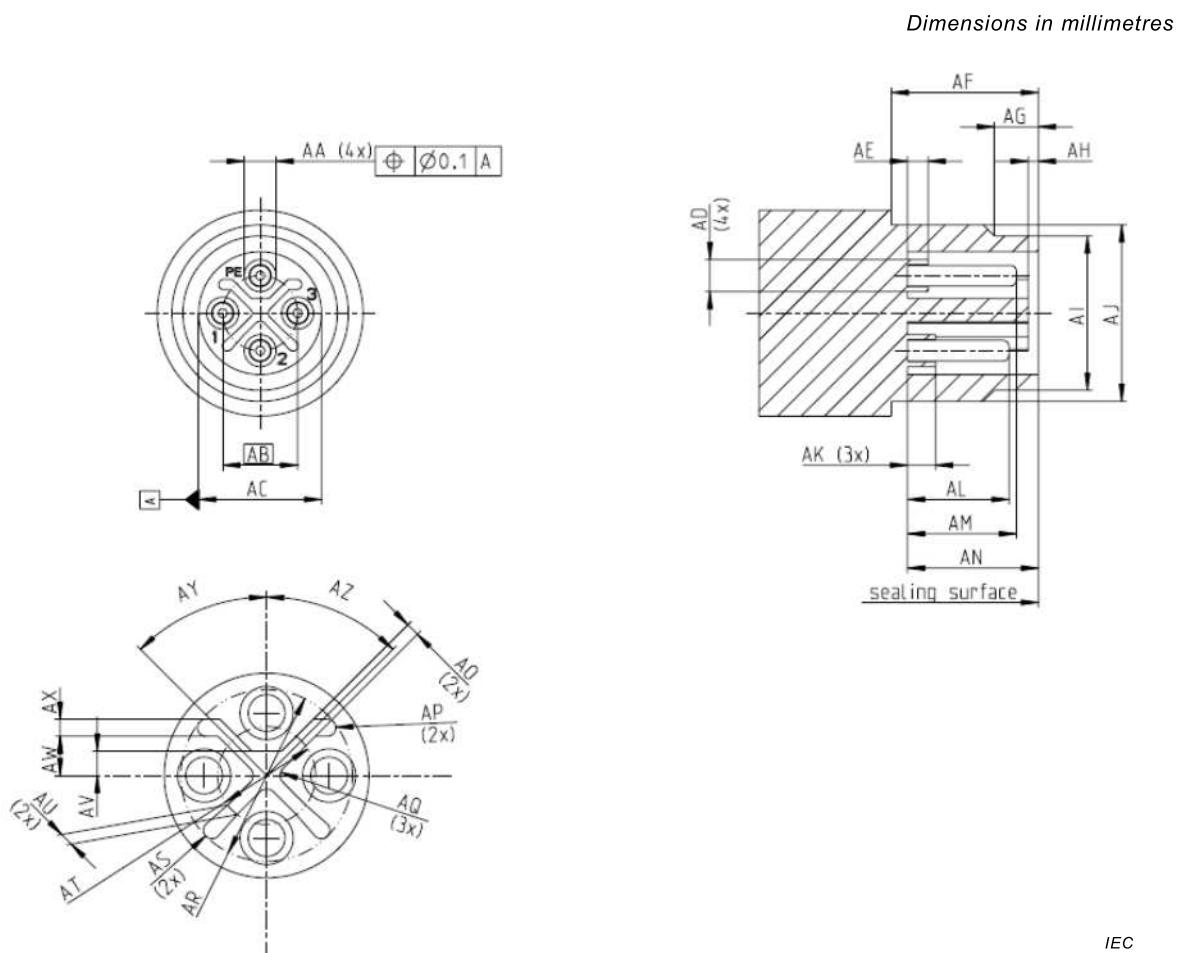


Figure 36 – S-coding male side

Table 23 shows the dimensions for Figure 36.

Table 23 – Dimensions for Figure 36

Dimensions in millimetres

Letter	Minimum	Nominal	Maximum
AA	Ø 1,47		Ø 1,5
AB		Ø 5,1	
AC	Ø 8,35	Ø 8,4	Ø 8,45
AD	Ø 2,07	Ø 2,12	Ø 2,17
AE	1,4	1,45	1,5
AF	10		
AG			3
AH	0,5	0,6	0,7
AI	Ø 10,2	Ø 10,35	Ø 10,5
AJ		M12x1	
AK	1,9	1,95	2
AL	6,5	6,7	6,9
AM	7,4	7,6	7,8
AN	8,7	8,8	8,9
AO	0,45	0,5	0,55
AP	R 0,3		R 0,35
AQ	R 0,4	R 0,45	R 0,5
AR	Ø 6,9	Ø 7	Ø 7,1
AS	R 0,3		R 0,35
AT	Ø 4	Ø 4,1	Ø 4,2
AU	0,45	0,5	0,55
AV	0,95		1
AW	1,55		1,6
AX	0,6	0,65	0,7
AY	44,5°	45°	45,5°
AZ	44,5°	45°	45,5°

5.3.6.2 Female side

Figure 37 shows the S-coding female side.

Dimensions in millimetres

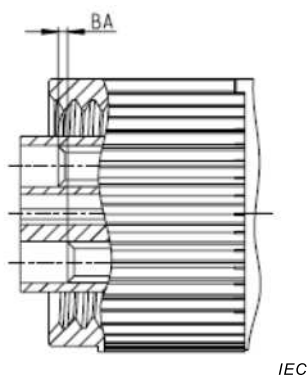


Figure 37 – S-coding female side

Table 24 shows the dimensions for Figure 37.

Table 24 – Dimensions for Figure 37

Dimensions in millimetres

Letter	Minimum	Nominal	Maximum
BA	0,5		

5.3.6.3 Contact position S-coding, front view

Figure 38 shows the contact position S-coding front view.

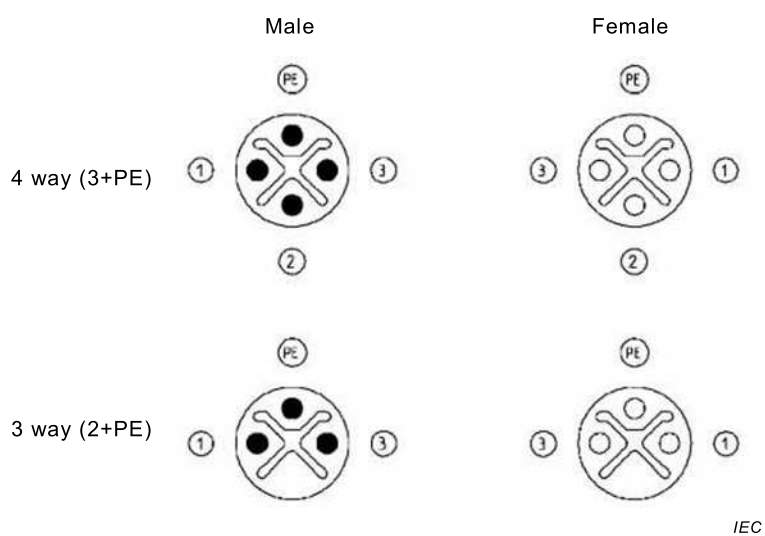


Figure 38 – Contact position S-coding front view

The contact marking shall be on the termination side of the connector insert as long as the size of the component allows the placement there.

5.3.7 T-coding

5.3.7.1 Male side

Figure 39 shows the T-coding male side.

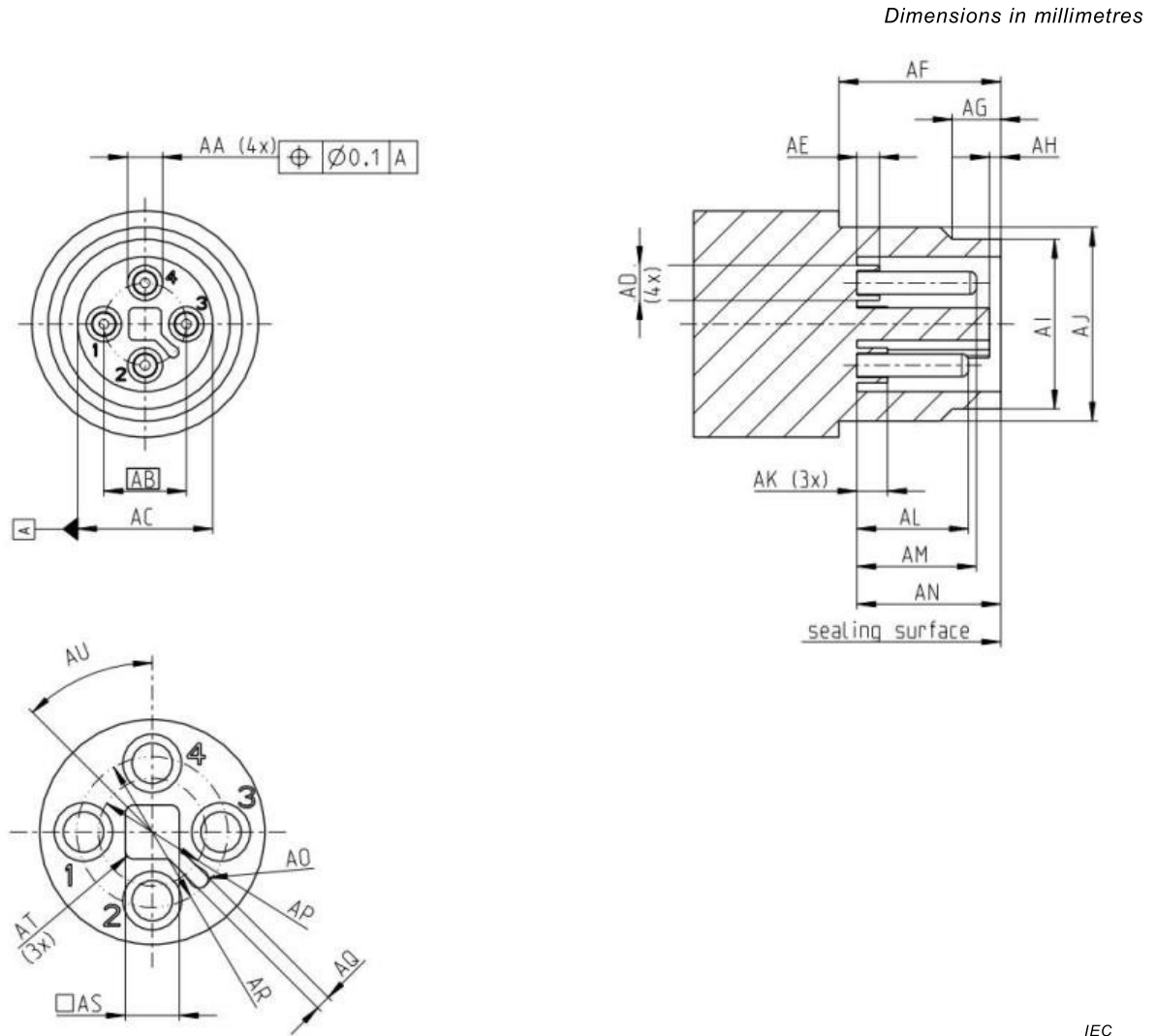


Figure 39 – T-coding male side

Table 25 shows the dimensions for Figure 39.

Table 25 – Dimensions for Figure 39

Dimensions in millimetres

Letter	Minimum	Nominal	Maximum
AA	Ø 1,47		Ø 1,5
AB		Ø 5,1	
AC	Ø 8,35	Ø 8,4	Ø 8,45
AD	Ø 2,07	Ø 2,12	Ø 2,17
AE	1,4	1,45	1,5
AF	10		
AG			3
AH	0,5	0,6	0,7
AI	Ø 10,2	Ø 10,35	Ø 10,5
AJ		M12x1	
AK	1,9	1,95	2
AL	6,5	6,7	6,9
AM	7,4	7,6	7,8
AN	8,7	8,8	8,9
AO	R 0,3		R 0,35
AP	Ø 4	Ø 4,1	Ø 4,2
AQ	0,45	0,5	0,55
AR	Ø 5,5	Ø 5,6	Ø 5,7
AS	1,9	1,95	2
AT	R 0,2	R 0,25	R 0,3
AU	44,5°	45°	45,5°

5.3.7.2 Contact position T-coding, front view

Figure 40 shows the contact position T-coding front view.

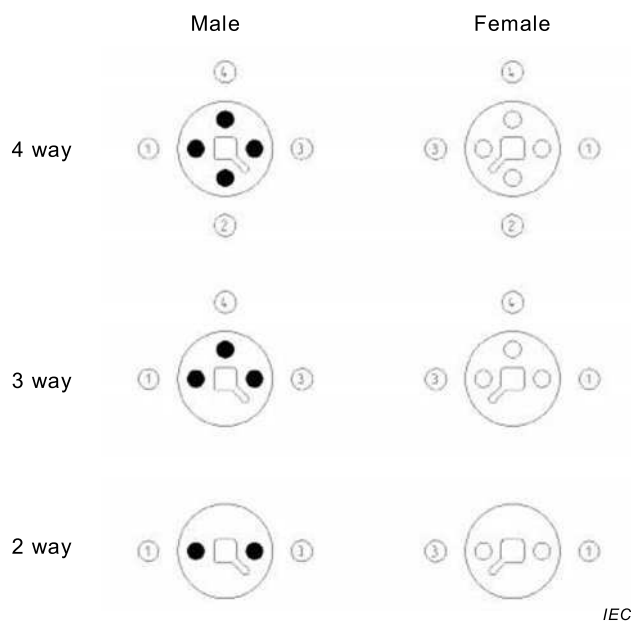


Figure 40 – Contact position T-coding front view

5.4 Engagement (mating) information

Arrows in Figure 41 indicate mating direction.

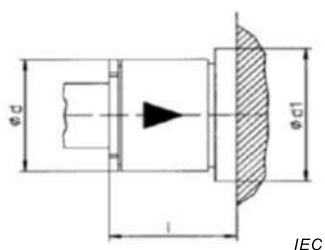


Figure 41a) – Housing variant 1

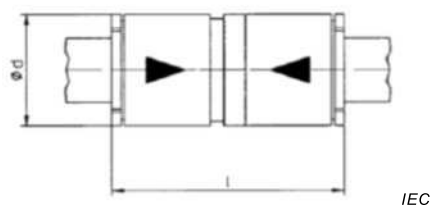


Figure 41b) – Housing variant 2

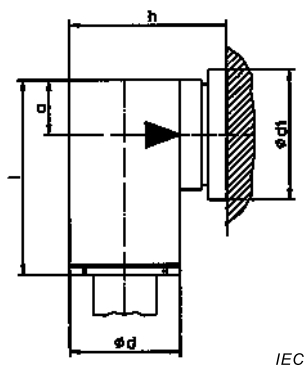


Figure 41c) – Housing variant 3

Figure 41 – Engagement (mating) information

Table 26 shows connectors dimensions in mated and locked position.

Table 26 – Connectors dimensions in mated and locked position

Dimensions in millimetres

Figure	Combination of styles	<i>a</i> max.	<i>d</i> max.	<i>d</i> 1	<i>h</i> ^a max.	<i>l</i> ^a max.
42a)	DM-JF	---	Ø 25	Ø 21	---	90
	DM-LF	---	Ø 18	Ø 21	---	83
	EM-JF	---	Ø 25	Ø 18,2	---	78
	EM-LF	---	Ø 18	Ø 18,2	---	71
	FM-JF	---	Ø 25	Ø 26,2	---	78
	FM-LF	---	Ø 18	Ø 26,2	---	71
	GM-JF	---	Ø 25	Ø 18,2	---	86
	GM-LF	---	Ø 18	Ø 18,2	---	79
	HM-JF	---	Ø 25	Ø 26,2	---	86
	HM-LF	---	Ø 18	Ø 26,2	---	79
	EF-JM	---	Ø 25	Ø 18,2	---	89
	EF-LM	---	Ø 18	Ø 18,2	---	82
	GF-JM	---	Ø 25	Ø 18,2	---	78
	GF-LM	---	Ø 18	Ø 18,2	---	73
	FF-JM	---	Ø 25	Ø 26,2	---	78
	FF-LM	---	Ø 18	Ø 26,2	---	73
	HF-JM	---	Ø 25	Ø 26,2	---	86
	HF-LM	---	Ø 18	Ø 26,2	---	81
	IM-JF	---	Ø 25	Ø 18,2	---	86
	IM-LF	---	Ø 18	Ø 18,2	---	81
	IF-JM	---	Ø 25	Ø 26,2	---	89
	IF-LM	---	Ø 18	Ø 26,2	---	84
42b)	JM-JF	---	Ø 25	---	---	132
	JM-LF	---	Ø 25	---	---	125
	JF-LM		Ø 25	---	---	127
	LM-LF		Ø 18	---	---	120

Figure	Combination of styles	<i>a</i> max.	<i>d</i> max.	<i>d</i> 1	<i>h</i> ^a max.	<i>l</i> ^a max.
42c)	DM-KF	13	Ø 25	Ø 21	70	53
	DM-MF	9	---	Ø 21	60	49
	EM-KF	13	Ø 25	Ø 18,2	58	53
	EM-MF	9	---	Ø 18,2	48	49
	FM-KF	13	Ø 25	Ø 26,2	58	53
	FM-MF	9	---	Ø 26,2	48	49
	GM-KF	13	Ø 25	Ø 18,2	66	53
	GM-MF	9	---	Ø 18,2	56	49
	HM-KF	13	Ø 25	Ø 26,2	66	53
	HM-MF	9	---	Ø 26,2	56	49
	IM-KF	13	Ø 25	Ø 18,2	69	53
	IM-MF	9	---	Ø 18,2	59	49
	EF-KM	13	Ø 25	Ø 18,2	58	53
	EF-MM	9	---	Ø 18,2	49	49
	FF-KM	13	Ø 25	Ø 26,2	58	52
	GF-KM	8	---	Ø 26,2	66	53
	GF-MM	11	Ø 18,2	Ø 26,2	57	49
	HF-KM	8	---	Ø 26,2	66	53
	HF-MM	11	Ø 25	Ø 26,2	57	49
	IF-KM	11	Ø 25	Ø 26,2	69	53
	IF-MM	8	Ø 18,2	Ø 26,2	60	49
a) Dimensions in mated and locked position, additional space for insertion: 15 mm.						

5.5 Gauges – Sizing gauges and retention force gauges

Figure 42 and Table 27 show the gauge dimensions.

Material: tool steel, hardened.

✓ =surface roughness according to ISO 1302: $Ra_{max} = 0,25 \mu m$, $Ra_{min} = 0,15 \mu m$ min.

Surface clean and free of grease.

Dimensions in millimetres

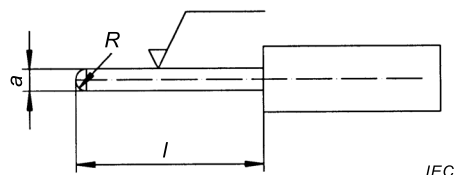


Figure 42 – Gauge dimensions

Table 27 shows gauges.

Table 27 – Gauges

Gauge	Mass g	Application	$\varnothing a$ mm	l min mm	Nom pin $\varnothing$ mm
P51	-	Sizing	$1,52^{+0,01}_0$	10	1,50
P52	25	Retention force	$1,48^0_{-0,01}$	10	
P53	-	Sizing	$1,02^{+0,01}_0$	10	1,00
P54	20	Retention force	$0,98^0_{-0,01}$	10	

6 Characteristics

6.1 General

Connectors shall be tested under the conditions given in this document and meet the relevant requirements specified. The methods according to IEC 62197-1 should be applied for quality assessment upon agreement between manufacturer and customer.

6.2 Contact assignment and other definitions

For contact assignment, see 5.3.

6.3 Classification into climatic category

Conditions: IEC 60068-1 and Table 28.

Table 28 – Climatic category

Climatic category	Category temperature		Damp heat steady state		Days
	Lower °C	Upper °C	Temperature °C	Relative humidity %	
25/085/-	-25	+85	40	93	-

6.4 Electrical characteristics

6.4.1 Creepage and clearance distances

Creepage and clearance distances shall be measured according to IEC 60512-1-2 with the following additional requirements.

For these connectors clearance and creepage distances shall be measured only in mated position. The minimum values for creepage distances can be found in Table 29 and clearance distances can be found in Table 30.

Table 29 – Creepage distances

Dimensions in millimetres

Coding	Number of contacts				
	2	3	4	5	6
E	-	2,0	2,0	2,0	-
F	2,0	2,0	2,0	-	-
K	-	6,3	6,3	6,3	-
L	-	2,0	2,0	2,0	-
M	-	6,3	6,3	6,3	6,3
S	-	6,3	6,3	-	-
T	2,0	2,0	2,0	-	-

Table 30 – Clearance distances

Dimensions in millimetres

Coding	Number of contacts				
	2	3	4	5	6
E	-	0,8	0,8	0,8	-
F	0,8	0,8	0,8	-	-
K	-	5,5	5,5	5,5	-
L	0,8	0,8	0,8	0,8	-
M	-	5,5	5,5	5,5	5,5
S	-	5,5	5,5	-	-
T	0,8	0,8	0,8	-	-

NOTE Coordination of insulation expressed by creepage distances and clearances provided by Tables 29 and 30 respectively is in accordance with IEC 60664-1 (and of course IEC 61984) for pollution degree 3, material group III, overvoltage category 3.

6.4.2 Voltage proof

Conditions: IEC 60512-4-1, test 4a, standard atmospheric conditions.

Mated connectors, Table 31 show voltage proof.

Table 31 – Voltage proof

kV r.m.s.

Coding	No. of contacts	Earth conductor contact	Between contacts		Between contacts and metal-housing	
			Fixed connectors	Free connectors	Fixed connectors	Free connectors
E	2	FE	0,84	0,84	0,84	0,84
	3					
	4					
F	2	Without	0,84	0,84	0,84	0,84
	3					
	4					
K	3	PE	3,31	3,31	3,31	3,31
	4					
	5					
L	3	FE	0,84	0,84	0,84	0,84
	4					
	5					
	2	Without				
	3					
	4					
M	3	PE	3,31	3,31	3,31	3,31
	4					
	5					
	6					
S	3	PE	3,31	3,31	3,31	3,31
	4					
T	2	without	0,84	0,84	0,84	0,84
	3					
	4					

NOTE Voltage proof values in Table 31 are in agreement with the requirements of IEC 60664-1 and IEC 61984.

6.4.3 Rated voltage – Rated impulse voltage – Pollution degree

Conditions: IEC 60664-1.

The permissible rated voltage depends on the application or specified safety requirement. Reductions in creepage or clearance distances may occur due to the printed board or wiring used and shall be duly taken into account. Table 32 shows rated voltage, rated impulse voltage and pollution degree.

Table 32 – Rated voltage – Rated impulse voltage – Pollution degree

Coding	No. of ways	Earth conductor contact	Rated voltage V	Rated impulse voltage kV	Pollution degree ^a
E	3	FE	63	1,5	3
	4				
	5				
F	2	without	63	1,5	3
	3				
	4				
K	3	PE	630	6	3
	4				
	5				
L	3	FE	63	1,5	3
	4				
	5				
	2	without			
	3				
	4				
M	3	PE	630	6	3
	4				
	5				
	6				
S	3	PE	630	6	3
	4				
T	2	without	63	1,5	3
	3				
	4				
^a Only in mated and locked condition.					

6.4.4 Current-carrying capacity

Conditions: IEC 60512-5-1, test 5a.

All contacts, but without PE contacts.

Rated current, see Table 1.

Values at 40 °C.

6.4.5 Contact resistance

Conditions: IEC 60512-2-1, test 2a.

Standard atmospheric conditions, measuring points, see 7.1.3 and Figure 43.

6.4.6 Insulation resistance

Conditions: IEC 60512-3-1, test 3a, Method A.

Standard atmospheric conditions, test voltage 500 V ± 15 V d.c.

6.4.7 Impedance

Not applicable (connectors not intended for data transmission).

6.5 Mechanical characteristics

6.5.1 Mechanical operation

Conditions: IEC 60512-9-1, test 9a.

Standard atmospheric conditions, speed of operations = 10 mm/s, rest: 30 s, unmated.

Table 33 shows the number of mechanical operations.

Table 33 – Performance level and number of mechanical operations

Performance level	Mechanical operations
PL1	100
PL2	50
PL3	20
other types	a)
a) Other mating cycles are upon agreement between manufacturer and user.	

6.5.2 Effectiveness of connector coupling devices

Not applicable (M12 screw locking is effective by design).

6.5.3 Insertion and withdrawal forces

Conditions: IEC 60512-13-2, test 13b.

Standard atmospheric conditions, speed = 10 mm/s.

Table 34 shows insertion and withdrawal forces requirements.

Table 34 – Insertion and withdrawal forces

Force in N

Number of poles	Total insertion force	Total withdrawal force
2 to 4	30 max.	30 max.
5 to 6	45 max.	45 max.

6.5.4 Contact retention in insert

Not applicable.

NOTE For removable crimp type contacts, upon agreement between manufacturer and customer.

6.5.5 Polarizing method

Conditions: IEC 60512-13-5, test 13e, standard atmospheric conditions.

Table 35 shows the polarizing insertion force.

Table 35 – Polarizing insertion forces

Number of poles	Total insertion force
2 to 4	1,5 x mating force, but not less than 35 N
5 to 6	1,5 x mating force, but not less than 40 N

6.6 Other characteristics

6.6.1 Vibration (sinusoidal)

Conditions: IEC 60512-6-4, test 6d, standard atmospheric conditions, connectors in mated and locked position.

The fixed and free connector shall be rigidly installed in a suitable fixture as specified in 7.1.4.

Vibration severity: frequency range 10 Hz to 500 Hz, displacement 0,35 mm peak-to-peak, acceleration 50 m/s² r.m.s.

Sweep cycle: 10 cycle on each of the 3 axes.

6.6.2 Shock

Conditions: IEC 60512-6-3, test 6c, acceleration 500 m/s².

The fixed and free connector shall be rigidly installed in a suitable fixture as specified in 7.1.4.

Shock severity: 11 ms, half-sine wave, 3 shocks in each axis and direction of axes, total 18 shocks.

6.6.3 Degree of protection provided by enclosures (IP-code)

Conditions: IP65 or IP65/IP67 according to IEC 60529, when connectors are in mated and locked position.

IP68 and/or IP69 upon agreement between manufacturer and user.

6.6.4 Screen and shielding properties

Not applicable.

6.7 Enviromental aspects – Marking of insulation material (plastics)

If applicable and reasonable, all plastic material shall be marked according to ISO 11469 to ease recycling.

7 Test schedule

7.1 General

7.1.1 Overview

This test schedule shows the tests and the order in which they shall be carried out as well as the requirements to be met. Unless otherwise specified, all tests shall be carried out under standard atmospheric conditions for testing as specified in IEC 60068-1 and as specified by the applicable part of IEC 60512.

Unless otherwise specified, mated and locked sets of connectors shall be tested. Care shall be taken to keep a particular combination of connectors during the complete test sequence, i.e. when unmating is necessary for a certain test, the same connector styles as before shall be mated for the subsequent tests.

In the following, a mated and locked set of connector styles is called a specimen.

When the initial tests P have been completed, the specimens are divided in the 4 test groups AP, BP, CP, and DP. In addition, 20 single contacts are used for EP.

Before testing commences, the connectors shall be stored for at least 24h in the non-engaged state under standard atmospheric conditions as per IEC 60068-1.

The necessary specimens are stated in Table 36.

Table 36 – Number of test specimens

	Test group					
	P	AP	BP	CP	DP	EP
No. of specimen	12	3	3	3	3	20 single contacts

7.1.2 Climatic category

Conditions: see 6.3.

7.1.3 Creepage and clearance distances

Conditions: see 6.4.1.

7.1.4 Arrangement for contact resistance measurements

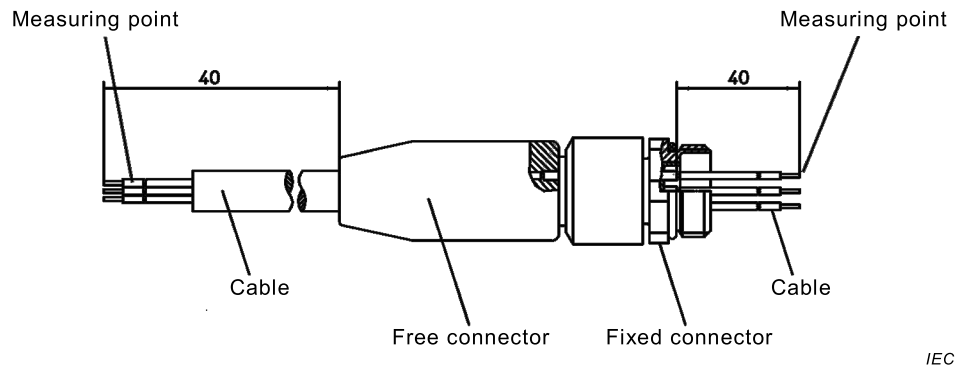
Conditions: see 6.4.5.

The measurement of contact resistance shall be carried out on the number of contacts specified. Any subsequent measurements of contact resistance shall be made on the same contacts.

Figure 43 shows the contact resistance arrangement.

The resistance of the wires between each measurement point as detailed in Figure 43 and the termination points between wire and contact on the free and fixed connectors should be measured or calculated.

The resistance of the wires shall be subtracted from the measured value to fulfil the specified contact.



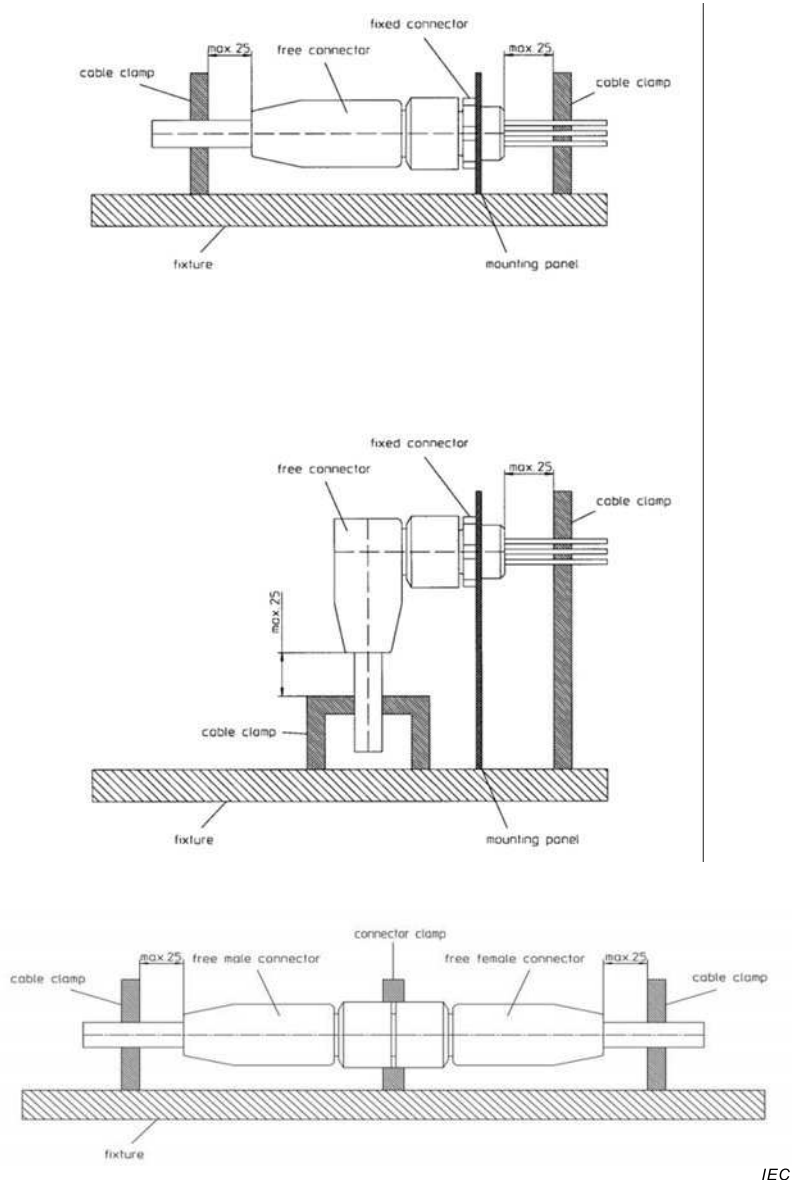
IEC

Figure 43 – Contact resistance arrangement

7.1.5 Arrangement for dynamic stress tests (vibration)

Conditions: see 6.6.1.

Figure 44 shows a dynamic stress test arrangement.



IEC

Figure 44 – Dynamic stress test arrangement**7.1.6 Arrangement for testing static load; axial**

Not applicable.

7.1.7 Wiring of specimens

As specified by the manufacturer and customer and representing relevant applications.

7.2 Test schedule**7.2.1 Test group P – Preliminary**

All specimens shall be subject to the tests of Table 37.

Table 37 – Test group P

Test phase	Test			Measurement to be performed		Requirements
	Title	IEC 60512 Test No.	Severity or condition of test	Title	IEC 60512 Test No.	All connector styles
P1	General examination	1	Unmated connectors	Visual Examination	1-1	There shall be no defect that would impair normal operation
				Dimensional examination	1-2	The dimensions shall comply with those specified in the relevant figure of Clause 5 and 6.4.1.
P2				Contact resistance	2-1	Shall be as per 6.4.5, all contacts, initial value: 5 mΩ max.
P3				Insulation resistance	3-1	Shall be as per 6.4.6, 100 MΩ min.
P4				Voltage proof	4-1	According to 6.4.2 No break down or flashover

The specimen shall be divided into groups as detailed in Table 36. All connectors in each group shall undergo the tests specified for the relevant group.

7.2.2 Test group AP – Dynamic/ Climatic

Table 38 shows the test group AP.

Table 38 – Test group AP

Test phase	Test			Measurement to be performed		Requirements
	Title	IEC 60512 Test No.	Severity or condition of test	Title	IEC 60512 Test No.	All connector styles
AP1				Insertion and withdrawal forces	13-2	Shall be as per 6.5.3 F_{max} see Table 34.
AP2			Female contacts only 3 contacts/specimen sizing and retention force gauge, see 5.5.1	Gauge retention force	16-5	Gauge P52 or P54 shall be retained as applicable.
AP3	Vibration	6-4	Sweeping cycles: 10, Full duration: 6 h	Contact disturbance	2-5	Shall be as per 6.6.1, duration of disturbance 1 μ s max.
				Contact resistance	2-1	Shall be as per 6.4.5, all contacts, rise in relation to initial values 10 m Ω max.
				Visual examination	1-1	There shall be no defect that would impair normal operation
AP4	Shock	6-3	3 shocks per direction of the 3 axes 18 shocks total	Contact disturbance	2-5	Shall be as per 6.6.2, Duration of disturbance 1 μ s max.
				Contact resistance	2-1	Shall be as per 6.4.5, all contacts, rise in relation to initial values 10 m Ω max.
				Visual examination	1-1	There shall be no defect that would impair normal operation
AP5	Rapid change of temperature	11-4	–25 °C to 85 °C, t = 30 min, 5 cycles	Contact resistance	2-1	Shall be as per 6.4.5, all contacts, rise in relation to initial values 10 m Ω max.
				Insulation resistance	3-1	Shall be as per 6.4.6, 100 M Ω min.
				Voltage proof	4-1	According to 6.4.2, no break down or flashover
				Visual examination	1-1	There shall be no defect that would impair normal operation
AP6	Climatic sequence	11-1				
AP6.1	Dry heat	11-9	Temp.: 85 °C Duration: 16 h	Insulation resistance at high temperature	3-1	Shall be as per 6.4.6, 100 M Ω min.
AP6.2	Damp heat, cyclic, first cycle	11-12	Method Db Temp.: 40 °C Recovery time: 2 h	Visual examination	1-1	There shall be no defect that would impair normal operation
AP6.3	Cold	11-10	Temp.: –25 °C Duration: 2 h Recovery time: 2 h	Visual examination	1-1	There shall be no defect that would impair normal operation
AP6.4	Damp heat, cyclic, remaining cycles	11-12	Conditions according to AP6.2 5 cycles Recovery time: 2 h	Contact resistance	2-1	Shall be as per 6.4.5, all contacts, rise in relation to initial values 10 m Ω max.
				Insulation resistance	3-1	Shall be as per 6.4.6, 100 M Ω min.

Test phase	Test			Measurement to be performed		Requirements
	Title	IEC 60512 Test No.	Severity or condition of test	Title	IEC 60512 Test No.	All connector styles
				Voltage proof	4-1	According to 6.4.2, no break down or flashover
				Insertion and withdrawal forces	13-2	Shall be as per 6.5.3
				Visual examination	1-1	There shall be no defect that would impair normal operation
AP7	IP Protection degree					
AP7.1	IP code second characteristic numeral		Conditions according to IEC 60529	IEC 60529:1989, Test 14.2.5 and Test 14.2.7		No leakage on contacts
AP7.2	IP code first characteristic numeral		Conditions according to IEC 60529:1989, Dust IP6X Test 6, table 7	IEC 60529:1989, Table 7		IP6X no deposit of dust on contacts ^a
AP7.3				Contact resistance	2-1	Shall be as per 6.4.5, all contacts, rise in relation to initial values 10 mΩ max.
				Insulation resistance	3-1	Shall be as per 6.4.6, 100 MΩ min.
				Voltage proof	4-1	According to 6.4.2, no break down or flashover
				Insertion and withdrawal forces	13-2	Shall be as per 6.5.3
AP8				Visual examination	1-1	There shall be no defect that would impair normal operation
AP9	Polarizing method	13-5	Shall be as per 6.5.5			It shall be possible to correctly align and mate the appropriate mating connectors. It shall not be possible to mate the connectors in any other than the correct manner
^a It is allowed to perform AP7.2 with an additional specimen, extending the total number of specimen by 1.						

7.2.3 Test group BP – Mechanical endurance

Table 39 shows test group BP.

Table 39 – Test group BP

Test phase	Test			Measurement to be performed		Requirements
	Title	IEC 60512 Test No.	Severity or condition of test	Title	IEC 60512 Test No.	
BP1			Female contacts only 3 contacts/specimen sizing and retention force gauge see 5.5	Gauge retention force	16-1	See 5.5.1, gauge P52 or P54 shall be retained as applicable.
BP2	Mechanical operation (half of the specified number of operations)	9-1	Shall be as per 6.5.1, performance level Table 33. Locking mechanism inoperative during operation.			
				Contact resistance	2-1	Shall be as per 6.4.5, all contacts, rise in relation to initial values 10 mΩ max.
				Visual examination	1-1	There shall be no defect that would impair normal operation
BP3	Climatic test					
BP3.1	Flowing mixed gas corrosion test	11-7	Flowing mixed gas corrosion – 4 days method 4 according to IEC 60068-2-60 samples in mated state	Contact resistance	2-1	Shall be as per 6.4.5, all contacts, rise in relation to initial values 10 mΩ max.
BP4	Mechanical operation (remaining half of specified number of operations)	9-1	Shall be as per 6.5.1, performance level Table 33. Locking mechanism inoperative during operation.	Contact resistance	2-1	Shall be as per 6.4.5, all contacts, rise in relation to initial values 10 mΩ max.
				Insulation resistance	3-1	Shall be as per 6.4.6, 100 MΩ min.
				Voltage proof	4-1	According to 6.4.2, no break down or flashover or flashover
				Visual examination	1-1	There shall be no defect that would impair normal operation
BP5.1	IP Protection degree, second characteristic numeral		Conditions according to IEC 60529	IEC 60529:1989, Test 14.2.5 and Test 14.2.7		No leakage on contacts
BP5.2				Insulation resistance	3-1	Shall be as per 6.4.6, 100 MΩ min.
				Voltage proof	4-1	According to 6.4.2 No break down or flashover
BP6				Insertion and withdrawal forces	13-2	Shall be as per 6.5.3
BP7			Female contacts only 3 contacts/specimen sizing and retention force gauge see 5.5	Gauge retention force	16-5	Shall be as per 5.5

7.2.4 Test group CP – Electrical load

Table 40 shows test group CP.

Table 40 – Test group CP

Test phase	Test			Measurement to be performed		Requirements
	Title	IEC 60512 Test No.	Severity or condition of test	Title	IEC 60512 Test No.	All connector styles
CP1	Rapid change of temperature	11-4	–25 °C to + 85 °C, t = 30 min, 5 cycles	Contact resistance	2-1	Shall be as per 6.4.5, all contacts, rise in relation to initial values 10 mΩ max.
				Insulation resistance	3-1	Shall be as per 6.4.6, 100 MΩ min.
				Voltage proof	4-1	According to 6.4.2 No break down or flashover
CP2	Mechanical operation	9-1	Shall be as per 6.5.1, performance level Table 33. Locking mechanism inoperative during operation.			
CP3	Electrical load and temperature	9-2	Duration: 1 000 h Ambient temp.: 40 °C Current load acc. to 5.2.3 Recovery time: 2 h Temp. sensor in centre of specimen	Contact resistance	2-1	Shall be as per 6.4.5, all contacts, rise in relation to initial values 10 mΩ max.
				Insulation resistance	3-1	Shall be as per 6.4.6, 100 MΩ min.
				Voltage proof	4-1	According to 6.4.2 No break down or flashover
CP4	IP Protection degree					
CP4.1	IP code second characteristic		Conditions according to IEC 60529	IEC 60529: 1989, Test 14.2.5 and Test 14.2.7		No leakage on contacts
CP4.2				Insulation resistance	3-1	Shall be as per 6.4.6, 100 MΩ min.
				Voltage proof	4-1	According to 6.4.2, no break down or flashover
CP5		1-1	Unmated connectors	Visual examination	1-1	There shall be no defect that would impair normal operation

7.2.5 Test group DP – Chemical resistivity

Table 41 shows a test group DP.

Table 41 – Test group DP

Test phase	Test			Measurement to be performed		Requirement
	Title	IEC 60512 Test No.	Severity or condition of test	Title	IEC 60512 Test No.	All connector styles
DP1	Resistance to fluids	19-3	5 cycles			The fluid used for testing is upon agreement between manufacturer and user.
DP2	Retreatment		Clearing of specimen by washing briefly in light petrol	Contact resistance	2-1	Shall be as per 6.4.5, all contacts, rise in relation to initial values 10 mΩ max.
DP3				Voltage proof	4-1	According to 6.4.2, no break down or flashover
DP4			Unmated connectors	Visual examination	1-1	There shall be no defect that would impair normal operation

7.2.6 Test group EP – Connection method tests

Table 42 shows the test group EP.

Table 42 – Test group EP

Test phase	Test			Measurement to be performed		Requirement
	Title	IEC 60512 Test No.	Severity or condition of test	Title	IEC 60512 Test No.	All connector styles
EP1	Solderless connections: screw, crimp, insulation displacement, insulation piercing, press-in	IEC 60352	See relevant IEC 60352 standard, for screw-type terminations see relevant IEC 60999-1			

Annex A (informative)

Diameter of the female connector body and orientation of coding

A.1 Diameter of the female connector body

For improvement of the coding safety between the different coding variants (coding types) A, B, C, D, E, F, and P, an outside diameter of the socket body of $\varnothing 8,3 \begin{smallmatrix} 0 \\ -0,1 \end{smallmatrix}$ is recommended, see Figure A.1 and Table A.1.

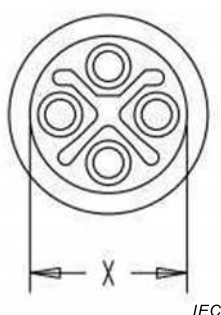


Figure A.1 – Diameter of the female connector body

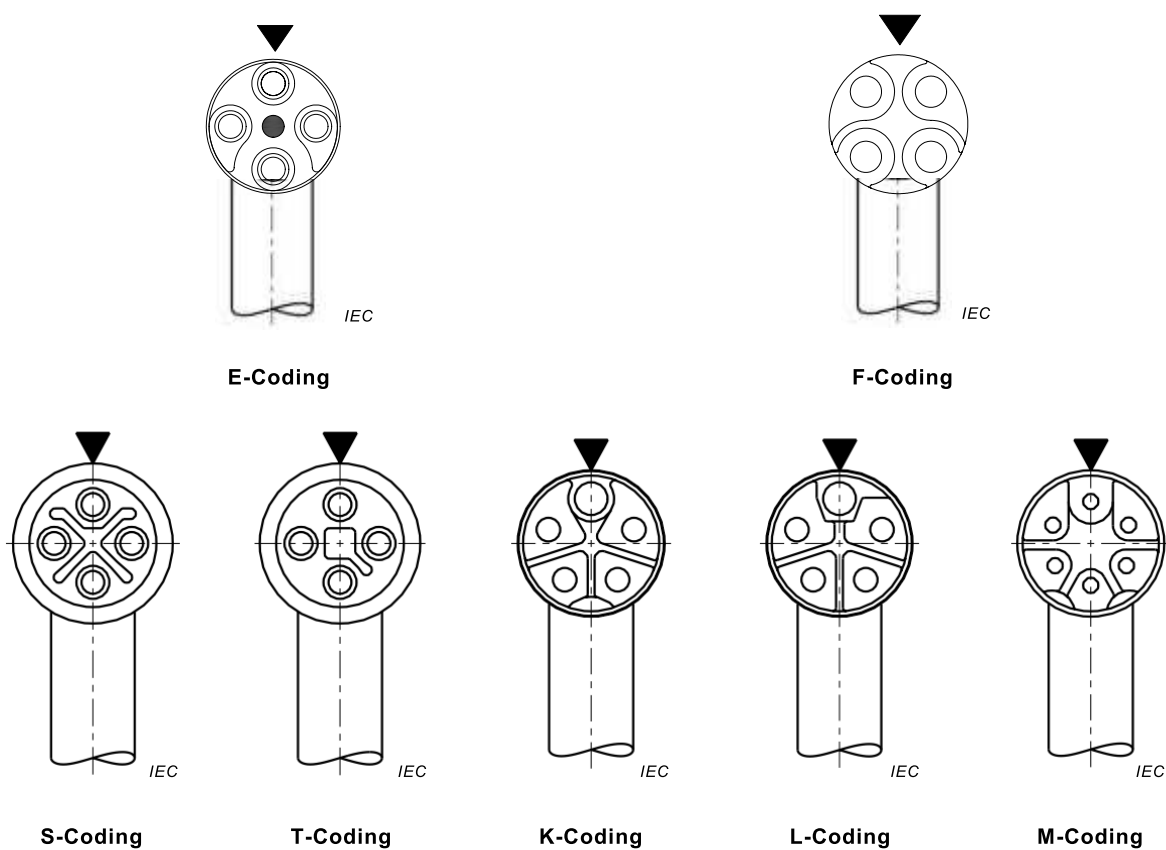
Table A.1 – Diameter of the female connector body,
dimension x, coding E, F, K, L, M, S, and T

Dimensions in mm

Coding variants	Dimension X
E	$\varnothing 9,2 \begin{smallmatrix} 0 \\ -0,1 \end{smallmatrix}$
F	$\varnothing 9,2 \begin{smallmatrix} 0 \\ -0,1 \end{smallmatrix}$
K	$\varnothing 8,75 \begin{smallmatrix} 0 \\ -0,1 \end{smallmatrix}$
L	
M	
S	$\varnothing 8,3 \begin{smallmatrix} 0 \\ -0,1 \end{smallmatrix}$
T	

A.2 Orientation of coding

For free connectors with right angled cable outlets, see 5.2.4 style KM, MM, KF and MF, the direction of the coding in relation to the cable outlet shall be specified to ensure that the cable leaves the housing in the specified direction, see Figure A.2.



**Figure A.2 – Orientation of cable outlet in relation to the coding –
Free male connectors according to Table 4**

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